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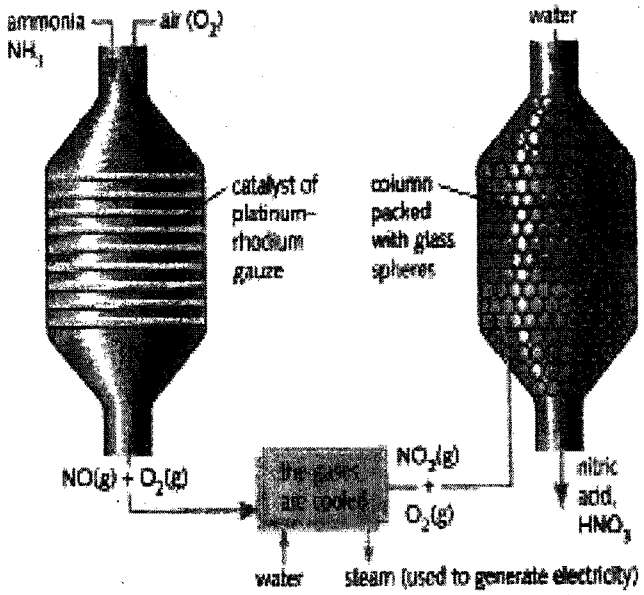
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Department of Examinations - Sri Lanka

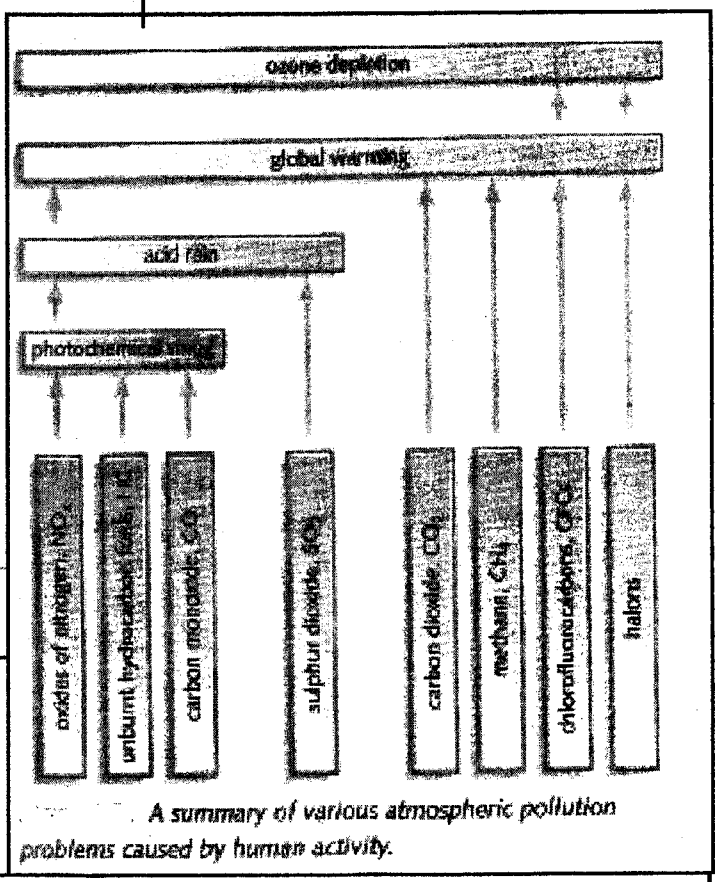
G.C.E. (A/L) Examination - 2018

02 - Chemistry

Marking Scheme



Producing nitric acid by the oxidation of ammonia.



A summary of various atmospheric pollution problems caused by human activity.

This document has been prepared for the use of Marking Examiners. Some changes would be made according to the views presented at the Chief Examiners' meeting.

Amendments to be included

G.C.E. (A/L) Examination - 2018

02 - Chemistry

Distribution of Marks

Paper I	:	1 X 50	=	50
Paper II	:			
Part A	:	100 X 4	=	400
Part B	:	150 X 2	=	300
Part C	:	150 X 2	=	300
Total			=	1000
Paper II - Final Marks			=	100

උත්තරපත්‍ර ලකුණු කිරීමේ පොදු ශිල්පීය ක්‍රම

උත්තරපත්‍ර ලකුණු කිරීමේ හා ලකුණු ලැයිස්තුවල ලකුණු සටහන් කිරීමේ සම්මත ක්‍රමය අනුගමනය කිරීම අනිවාර්යයෙන් ම කළ යුතුවේ. ඒ සඳහා පහත පරිදි කටයුතු කරන්න.

1. උත්තරපත්‍ර ලකුණු කිරීමට රතුපාට බෝල් පොයින්ට් පෑනක් පාවිච්චි කරන්න.

2. සෑම උත්තරපත්‍රයකම මුල් පිටුවේ සහකාර පරීක්ෂක සංකේත අංකය සටහන් කරන්න.

   ඉලක්කම් ලිවීමේදී **පැහැදිලි ඉලක්කමෙන්** ලියන්න.

3. ඉලක්කම් ලිවීමේදී වැරදුණු අවස්ථාවක් වේ නම් එය පැහැදිලිව තනි ඉරකින් කපා හැර නැවත ලියා කෙටි අත්සන යොදන්න.

4. එක් එක් ප්‍රශ්නයේ අනු කොටස්වල පිළිතුරු සඳහා හිමි ලකුණු ඒ ඒ කොටස අවසානයේ △ ක් තුළ ලියා දක්වන්න. අවසාන ලකුණු ප්‍රශ්න අංකයක් සමඟ □ ක් තුළ, භාග සංඛ්‍යාවක් ලෙස ඇතුළත් කරන්න. ලකුණු සටහන් කිරීම සඳහා පරීක්ෂකවරයාගේ ප්‍රයෝජනය සඳහා ඇති තීරුව භාවිත කරන්න.

උදාහරණ : ප්‍රශ්න අංක 03

(i) ..... √ △ —

(ii) ..... √ △ 3/5

(iii) ..... √ △ 3/5

03 (i) $\frac{4}{5}$ + (ii) $\frac{3}{5}$ + (iii) $\frac{3}{5}$ = $\boxed{\frac{10}{15}}$

බහුවරණ උත්තරපත්‍ර : (කවුළු පත්‍රය)

1. අ.පො.ස. (උ.පෙළ) හා තොරතුරු තාක්ෂණ විභාගය සඳහා කවුළු පත්‍ර දෙපාර්තමේන්තුව මගින් සකසනු ලැබේ. නිවැරදි වරණ කපා ඉවත් කළ සහතික කරන ලද කවුළුපතක් ඔබ වෙත සපයනු ලැබේ. සහතික කළ කවුළු පත්‍රයක් භාවිත කිරීම පරීක්ෂකගේ වගකීම වේ.

2. අනතුරුව උත්තරපත්‍ර හොඳින් පරීක්ෂා කර බලන්න. කිසියම් ප්‍රශ්නයකට එක් පිළිතුරකට වඩා ලකුණු කර ඇත්නම් හෝ එකම පිළිතුරක්වත් ලකුණු කර නැත්නම් හෝ වරණ කැපී යන පරිදි ඉරක් අඳින්න. ඇතැම් විට අයදුම්කරුවන් විසින් මුලින් ලකුණු කර ඇති පිළිතුරක් මකා වෙනත් පිළිතුරක් ලකුණු කර තිබෙන්නට පුළුවන. එසේ මකන ලද අවස්ථාවකදී පැහැදිලිව මකා නොමැති නම් මකන ලද වරණය මත ද ඉරක් අඳින්න.

3. කවුළු පත්‍රය උත්තරපත්‍රය මත නිවැරදිව තබන්න. නිවැරදි පිළිතුර ✓ ලකුණකින් ද, වැරදි පිළිතුර 0 ලකුණකින් ද වරණ මත ලකුණු කරන්න. නිවැරදි පිළිතුරු සංඛ්‍යාව ඒ ඒ වරණ තීරයට පහළින් ලියා දක්වන්න. අනතුරුව එම සංඛ්‍යා එකතු කර මුළු නිවැරදි පිළිතුරු සංඛ්‍යාව අදාළ කොටුව තුළ ලියන්න.

ව්‍යුහගත රචනා හා රචනා උත්තරපත්‍ර :

1. අයදුම්කරුවන් විසින් උත්තරපත්‍රයේ හිස්ව තබා ඇති පිටු හරහා රේඛාවක් ඇඳ කපා හරින්න. වැරදී හෝ නුසුදුසු පිළිතුරු යටින් ඉරි අදින්න. ලකුණු දිය හැකි ස්ථානවල හරි ලකුණු යෙදීමෙන් එය පෙන්වන්න.

2. ලකුණු සටහන් කිරීමේදී ඕවර්ලන්ඩ් කඩදාසියේ දකුණු පස තීරය යොදා ගත යුතු වේ.

3. සෑම ප්‍රශ්නයකටම දෙන මුළු ලකුණු උත්තරපත්‍රයේ මුල් පිටුවේ ඇති අදාළ කොටුව තුළ ප්‍රශ්න අංකය ඉදිරියෙන් අංක දෙකකින් ලියා දක්වන්න. ප්‍රශ්න පත්‍රයේ දී ඇති උපදෙස් අනුව ප්‍රශ්න තෝරා ගැනීම කළ යුතුවේ. සියලු ම උත්තර ලකුණු කර ලකුණු මුල් පිටුවේ සටහන් කරන්න. ප්‍රශ්න පත්‍රයේ දී ඇති උපදෙස්වලට පටහැනිව වැඩි ප්‍රශ්න ගණනකට පිළිතුරු ලියා ඇත්නම් අඩු ලකුණු සහිත පිළිතුරු කපා ඉවත් කරන්න.

4. පරීක්ෂාකාරීව මුළු ලකුණු ගණන එකතු කොට මුල් පිටුවේ නියමිත ස්ථානයේ ලියන්න. උත්තරපත්‍රයේ සෑම උත්තරයකටම දී ඇති ලකුණු ගණන උත්තරපත්‍රයේ පිටු පෙරළමින් නැවත එකතු කරන්න. එම ලකුණ ඔබ විසින් මුල් පිටුවේ එකතුව ලෙස සටහන් කර ඇති මුළු ලකුණට සමාන දැයි නැවත පරීක්ෂා කර බලන්න.

ලකුණු ලැයිස්තු සකස් කිරීම :

මෙවර සියලු ම විෂයන්හි අවසාන ලකුණු ඇගයීම් මණ්ඩලය තුළදී ගණනය කරනු නොලැබේ. එබැවින් එක් එක් පත්‍රයට අදාළ අවසාන ලකුණු වෙන වෙනම ලකුණු ලැයිස්තුවලට ඇතුළත් කළ යුතු ය. I පත්‍රයට අදාළ ලකුණු ලකුණු ලැයිස්තුවේ "I වන පත්‍රය" තීරුවේ ඇතුළත් කර අකුරෙන් ද ලියන්න. අදාළ විස්තර ලකුණු ඇතුළත් කර "II වන පත්‍රය" තීරුවේ II පත්‍රයේ අවසාන ලකුණු ඇතුළත් කරන්න. 51 චිත්‍ර විෂයයේ I, II හා III පත්‍රවලට අදාළ ලකුණු වෙන වෙනම ලකුණු ලැයිස්තුවල ඇතුළත් කර අකුරෙන් ද ලිවිය යුතු වේ.

***

සියලු ම හිමිකම් ඇවිරිණි / முழுப் பதிப்புரிமையுடையது / All Rights Reserved]

இலங்கைப் பரீட்சைத் திணைக்களம் Department of Examinations, Sri Lanka

අධ්‍යයන පොදු සහතික පත්‍ර (උසස් පෙළ) විභාගය, 2018 අගෝස්තු
கல்விப் பொதுத் தராதர பத்திர (உயர் தர)ப் பரீட்சை, 2018 ஓகஸ்ற்று
General Certificate of Education (Adv. Level) Examination, August 2018

15.08.2018 / 0830 – 1030

රසායන විද්‍යාව I
இரசாயனவியல் I
Chemistry I

02 E I

පැය දෙකයි
இரண்டு மணித்தியாலம்
Two hours

Instructions:

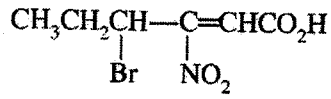
* *Periodic Table is provided.*
* *This paper consists of* **09** *pages.*
* *Answer* ***all*** *the questions.*
* ***Use of calculators is not allowed.***
* *Write your* ***Index Number*** *in the space provided in the answer sheet.*
* *Follow the instructions given on the back of the answer sheet carefully.*
* *In each of the questions* **1** *to* **50**, *pick one of the alternatives from* (1), (2), (3), (4), (5) ***which is correct or most appropriate and mark your response on the answer sheet with a cross*** (×) ***in accordance with the instructions given on the back of the answer sheet.***

Universal gas constant $R = 8.314\ \text{J K}^{-1}\ \text{mol}^{-1}$
Avogadro constant $N_A = 6.022 \times 10^{23}\ \text{mol}^{-1}$
Planck's constant $h = 6.626 \times 10^{-34}\ \text{J s}$
Velocity of light $c = 3 \times 10^{8}\ \text{m s}^{-1}$

1. The number of unpaired electrons present in a gaseous Co^{3+} ion in its ground state is,
(1) 1 (2) 2 (3) 3 (4) 4 (5) 5

2. Which quantum number(s) (n, l, m_l, m_s) is/are associated with the shape of an atomic orbital of an atom?
(1) l (2) m_l (3) n and l (4) n and m_l (5) l and m_l

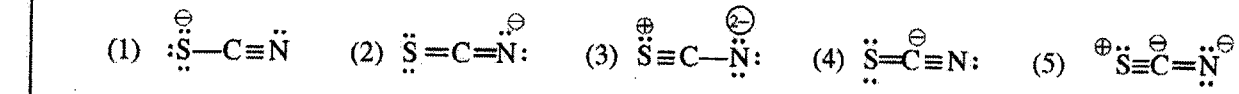
3. What is the IUPAC name of the compound shown below?



(1) 4–bromo–3–nitro–2–hexenoicacid
(2) 4–bromo–3–nitro–2–hexenoic acid
(3) 3–nitro–4–bromo–2–hexenoicacid
(4) 3–nitro–4–bromo–2–hexenoic acid
(5) 3–bromo–4–nitro–4–hexenoic acid

4. The **correct** answer when the molecules O_2, H_2O, H_2O_2, OF_2 and O_2F_2 (structure similar to H_2O_2) are arranged in the **decreasing** order of the oxidation state of oxygen (O) is,
(1) $O_2F_2 > OF_2 > O_2 > H_2O > H_2O_2$
(2) $H_2O > H_2O_2 > O_2 > O_2F_2 > OF_2$
(3) $H_2O_2 > O_2F_2 > O_2 > OF_2 > H_2O$
(4) $OF_2 > O_2F_2 > O_2 > H_2O > H_2O_2$
(5) $OF_2 > O_2F_2 > O_2 > H_2O_2 > H_2O$

5. The **most** acceptable Lewis structure for the thiocyanate ion SCN^- is,



6. The molarity (mol dm^{-3}) of a NaI solution which has a density of 1.03 g cm^{-3} and is 3% NaI by mass is,
(Na = 23, I = 127)
(1) 0.21 (2) 0.23 (3) 0.25 (4) 0.28 (5) 0.30

7. Precipitates of AgI and AgBr were added to a small amount of distilled water. This mixture was allowed to reach equilibrium at 25 °C. It was observed that both the solids were present in the system at equilibrium. Which of the following relations is applicable to this solution?
($K_{sp(AgI)} = 8.0 \times 10^{-17}$ mol^2 dm^{-6}, $K_{sp(AgBr)} = 5.0 \times 10^{-13}$ mol^2 dm^{-6} at 25 °C)

(1) $[Br^-] = \sqrt{5.0 \times 10^{-13}}$ mol dm^{-3} and $[I^-] = \sqrt{8.0 \times 10^{-17}}$ mol dm^{-3}

(2) $[Br^-]\,[I^-] = [Ag^+]^2$

(3) $\left[Ag^+\right] = \left(\sqrt{5.0 \times 10^{-13}} + \sqrt{8.0 \times 10^{-17}}\right)$ mol dm^{-3}

(4) $\frac{[Br^-]}{[I^-]} = \frac{5.0}{8.0} \times 10^4$

(5) $[Ag^+] = [Br^-] = [I^-]$

8. Which of the following statements is **false**?
(1) Although the carbonates of all the group two metals in the Periodic Table are insoluble in water, their bicarbonates are soluble.
(2) The hydroxides of all the group two metals in the Periodic Table are soluble in water.
(3) The nitrates of all the group two metals in the Periodic Table are soluble in water.
(4) The oxides and hydroxides of Na and Mg show basic properties whereas the oxide and hydroxide of Al show amphoteric properties.
(5) The hydrides of Si and S show weakly acidic properties.

9. In which list are the elements given in the order of **increasing** (left to right) atomic radii?
(1) Li, Na, Mg, S (2) C, Si, S, Cl (3) B, C, N, P
(4) Li, Na, K, Ca (5) B, Be, Na, K

10. Liquids A and B form an ideal solution. Consider a mixture of liquids A and B in equilibrium with the vapour in a closed rigid container at constant temperature. P_A^o and P_B^o respectively are the saturated vapour pressures of A and B while P is the total pressure of the container and X_A^g is the mole fraction of A in the vapour phase. Which of the following is correct about this system?

(1) $P = \left(P_A^o - P_B^o\right) X_A^g + P_B^o$ (2) $\frac{1}{P} = \left(\frac{1}{P_A^o} - \frac{1}{P_B^o}\right) X_A^g + \frac{1}{P_B^o}$

(3) $P = \left(P_A^o + P_B^o\right) X_A^g - P_B^o$ (4) $\frac{1}{P} = \left(\frac{1}{P_B^o} - \frac{1}{P_A^o}\right) \frac{1}{X_A^g}$

(5) $\frac{1}{P} = \left(\frac{1}{P_A^o} - \frac{1}{P_B^o}\right) \frac{1}{X_A^g}$

11. The **increasing** order of boiling points of the following substances is,
He, CH_4, CCl_4, CBr_4, SiH_4
(1) $CH_4 < He < SiH_4 < CCl_4 < CBr_4$ (2) $He < SiH_4 < CH_4 < CCl_4 < CBr_4$
(3) $He < CH_4 < SiH_4 < CCl_4 < CBr_4$ (4) $CH_4 < He < SiH_4 < CBr_4 < CCl_4$
(5) $He < CH_4 < CCl_4 < SiH_4 < CBr_4$

12. Identify the **correct** statement from the following.
(1) Among the electronic transitions $n=2 \longrightarrow n=1$, $n=3 \longrightarrow n=2$ and $n=4 \longrightarrow n=3$ in a hydrogen atom, most energy is released in $n=3 \longrightarrow n=2$.
(2) Among the species OF_2, OF_4 and SF_4, the least stable is SF_4.
(3) Among the elements Li, C, N, Na and P, the least electronegative element is Li.
(4) In the following pairs (Li & F), (Li^+ & F^-), (Li^+ & O^{2-}) and (O^{2-} & F^-), the difference in radii is greatest between Li^+ and O^{2-}.
(5) The only type of intermolecular force present in CH_2Cl_2 in the liquid phase is dipole-dipole forces.

13. Consider the reaction: $CH_4(g) \longrightarrow CH_3(g) + H(g)$

The standard change in enthalpy of the above reaction is,

(1) the standard enthalpy change for the dissociation of the first C—H bond in methane.
(2) the standard atomisation enthalpy change of methane.
(3) the standard first ionisation enthalpy change of methane.
(4) the standard bond dissociation enthalpy change of methane.
(5) the standard radical formation enthalpy change of methane.

14. The elementary reaction $2A(g) \longrightarrow B(g)$ occurs in a closed rigid container at a constant temperature. Initial pressure of the container is P_o and the pressure when the rate of reaction is 50% of the initial value is P_t. Which of the following gives the correct value for $\frac{P_t}{P_o}$?

(1) $\frac{P_t}{P_o} = \frac{1}{2}$ (2) $\frac{P_t}{P_o} = \frac{1}{\sqrt{2}}$ (3) $\frac{P_t}{P_o} = \frac{1+\sqrt{2}}{2\sqrt{2}}$ (4) $\frac{P_t}{P_o} = \frac{\sqrt{2}}{1+\sqrt{2}}$ (5) $\frac{P_t}{P_o} = \frac{\sqrt{2}-1}{1+\sqrt{2}}$

15. An equimolar aqueous solution of the weak acids HA and HB (1.0 mol dm^{-3} in each acid) with pK_a values 4.7 and 5.0 respectively is at equilibrium. The value of $\log\left(\frac{[A^-]}{[B^-]}\right)$ is approximately equal to,

(1) 23.5 (2) −0.3 (3) 0.3 (4) 0.94 (5) 1.06

16. Which of the following statements about C_6H_5OH is **false**?

(1) Reacts with CH_3COCl to form a phenyl ester.
(2) Reacts with bromine water to give a white precipitate.
(3) Evolves CO_2 gas when treated with $NaHCO_3$.
(4) Gives a coloured compound when treated with $C_6H_5N_2^+Cl^-$ in the presence of NaOH.
(5) Gives a coloured (purplish) solution when treated with neutral $FeCl_3$.

17. The half life of a reaction is,

(1) always independent of the initial concentration of reactants.
(2) always dependent on the rate constant.
(3) always independent of the order of the reaction.
(4) always independent of temperature.
(5) equal to twice the total reaction time.

18. Electromotive force of an electrochemical cell does **not** depend on,

(1) the nature of the electrolytes.
(2) temperature.
(3) the concentrations of the electrolytes.
(4) the surface areas of the electrodes.
(5) the types of metals that form the electrodes.

19. IO_3^- (iodate ion) oxidizes the SO_3^{2-} ion to SO_4^{2-} in acidic medium. The mass of KIO_3 required to totally oxidize the amount of Na_2SO_3 present in 25.0 cm^3 of a solution of Na_2SO_3 (0.50 mol dm^{-3}) to Na_2SO_4 is 1.07 g. (O = 16, K = 39, I = 127)

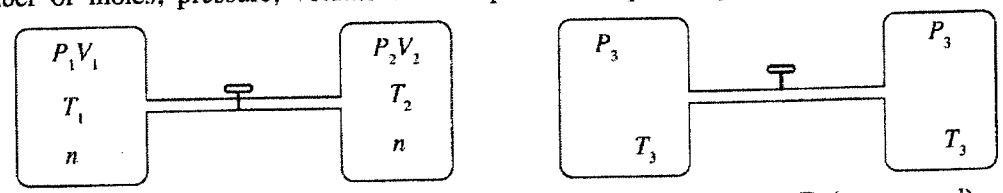
The final oxidation state of iodine after the completion of the reaction is,

(1) −1 (2) 0 (3) +1 (4) +2 (5) +3

20. Which of the following statements is **false** with regard to the *s*-block elements in the Periodic Table?

(1) All elements in group I react with water liberating H_2 gas.
(2) All elements in group I except Li react with N_2 gas.
(3) All elements in group II react with N_2 gas.
(4) Na reacts with excess O_2 to give Na_2O_2 whereas K gives KO_2.
(5) All elements in the *s*-block are good reducing agents.

21. A system consisting of two rigid containers containing an ideal gas is shown in the diagram. The containers can be connected to each other by opening the tap. The system changes from configuration **A** to configuration **B** when the tap is opened. In general n, P, V and T represent number of moles, pressure, volume and temperature respectively.



configuration **A** (tap closed) configuration **B** (tap opened)

Which of the following relations is **correct** about this system?

(1) $P_1V_1 = P_2V_2$ (2) $\frac{P_3T_1}{P_1} + \frac{P_3T_2}{P_2} = 2T_3$ (3) $\frac{T_1}{P_1} = \frac{T_2}{P_2}$

(4) $P_1T_1 = P_2T_2$ (5) $P_1V_1 + P_2V_2 = P_3(V_1 + V_2)$

22. Which of the following statements is **false** with regard to 3*d*-elements of the Periodic Table?

(1) Atomic radii are smaller than the atomic radii of the *s*-block elements in the same period.
(2) Densities are higher than the densities of the *s*-block elements in the same period.
(3) V_2O_5, CrO_3 and Mn_2O_7 are acidic oxides.
(4) First ionization energies are less than the first ionization energies of the *s*-block elements in the same period.
(5) The most common oxidation states of cobalt in cobalt compounds are +2 and +3.

23. Standard Gibbs energy changes for the reaction, $MO(s) \rightarrow M(s) + \frac{1}{2}O_2(g)$ at two different temperatures are given below.

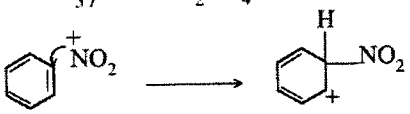
T/K	$\Delta G°$/kJ mol^{-1}
1000	–100.2
2000	–148.6

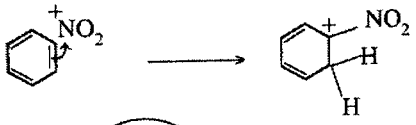
The standard entropy change of the reaction is,

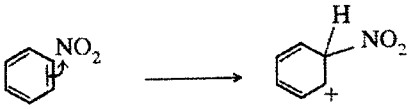
(1) 248.8 J K^{-1} mol^{-1} (2) –248.8 J K^{-1} mol^{-1} (3) –48.4 J K^{-1} mol^{-1}

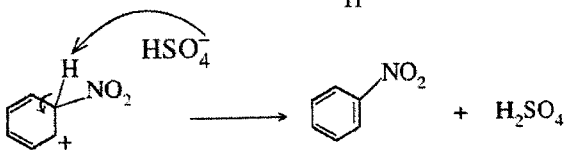
(4) 348.4 J K^{-1} mol^{-1} (5) 48.4 J K^{-1} mol^{-1}

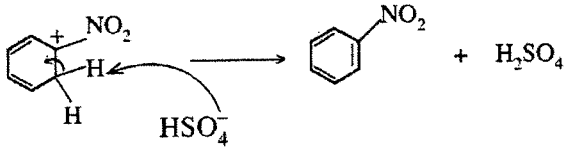
24. Which of the following represents a **correct** step in the mechanism of nitration of benzene with conc. HNO_3 / conc. H_2SO_4?

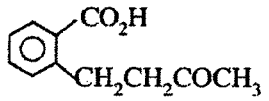
(1) 

(2) 

(3) 

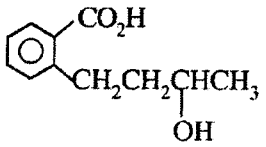
(4) 

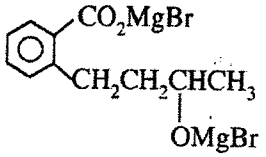
(5) 

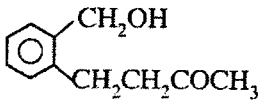
25. 

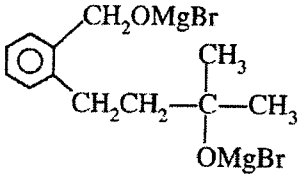
$\xrightarrow[(2)\ H^+/H_2O]{(1)\ NaBH_4}$ **X** $\xrightarrow[CH_3MgBr]{excess}$ **Y**

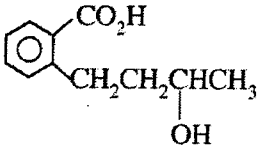
In the reaction sequence given above, the structures of **X** and **Y** respectively are,

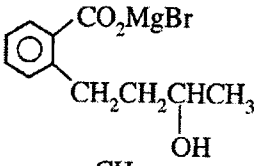
(1) 

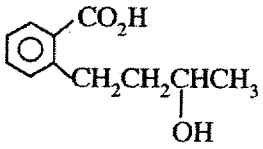
, 

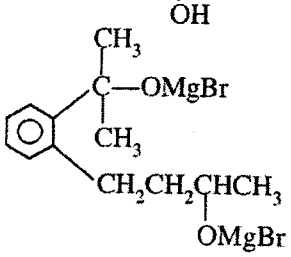
(2) 

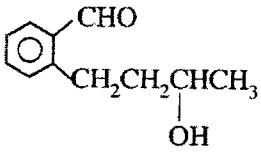
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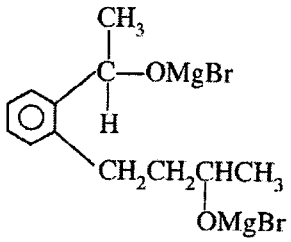
(3) 

, 

(4) 

, 

(5) 

, 

26. When $(NH_4)_2CO_3(s)$, $(NH_4)_2Cr_2O_7(s)$ and $NH_4NO_3(s)$ are heated, the nitrogen containing compounds obtained are respectively,

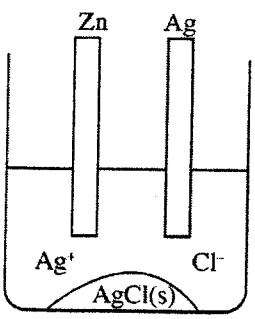
(1) NH_3, N_2 and NO_2 (2) N_2O, N_2 and NH_3 (3) NH_3, N_2 and N_2O
(4) N_2, N_2O and NH_3 (5) N_2, NH_3 and N_2O

27. Which of the following would occur soon after connecting a rod of Zn and a rod of Ag immersed in a saturated solution of AgCl and AgCl(s) kept in a beaker as shown in the diagram, by a conductor?

$Zn^{2+}(aq) + e \longrightarrow Zn(s)$ $E^\circ = -0.76$ V

$Ag^+(aq) + e \longrightarrow Ag(s)$ $E^\circ = 0.80$ V

(1) Zn dissolves, Ag deposits, AgCl(s) dissolves.
(2) Zn dissolves, Ag dissolves, AgCl(s) dissolves.
(3) Zn dissolves, Ag dissolves, AgCl(s) deposits.
(4) Zn deposits, Ag dissolves, AgCl(s) dissolves.
(5) Chloride concentration in the solution decreases.



28. In the reaction sequence given below, the structures of **P** and **Q** respectively are,

$$C_6H_5C{\equiv}CH \xrightarrow{Hg^{2+}/\text{dil. } H_2SO_4} \mathbf{P} \xrightarrow[\text{conc. HCl}]{Zn/Hg} \mathbf{Q}$$

(1) $C_6H_5\underset{\underset{OH}{|}}{C}{=}CH_2$, $C_6H_5CH{=}CH_2$

(2) $C_6H_5CH{=}\underset{\underset{OH}{|}}{C}H$, $C_6H_5CH{=}CH_2$

(3) $C_6H_5{-}\overset{\overset{O}{\|}}{C}{-}CH_3$, $C_6H_5{-}\underset{\underset{H}{|}}{\overset{\overset{OH}{|}}{C}}{-}CH_3$

(4) $C_6H_5{-}\overset{\overset{O}{\|}}{C}{-}CH_3$, $C_6H_5CH_2CH_3$

(5) $C_6H_5\underset{\underset{OH}{|}}{C}{=}CH_2$, $C_6H_5\underset{\underset{OH}{|}}{C}HCH_3$

29. Which of the following statements is **incorrect** regarding polymers?

(1) Bakelite is a thermosetting polymer.
(2) Teflon is a thermoplastic polymer.
(3) Nylon 6,6 is formed by addition polymerisation between 1,6–diaminohexane and hexanedioic acid.
(4) Terelene is formed by condensation polymerisation between ethylene glycol and terephthalic acid.
(5) Natural rubber consists of *cis*-polyisoprene chains.

30. An experiment was carried out to find the order (m) with respect to $S_2O_3^{2-}$ of the reaction $S_2O_3^{2-}(aq) + 2H^+(aq) \longrightarrow H_2O(l) + SO_2(g) + S(s)$. Initial rate of the reaction (R) was measured by adding different volumes (v) of 0.01 mol dm^{-3} $S_2O_3^{2-}$ into a solution of an acid. The H^+ concentration of the reaction mixture was kept constant, but the total volume (V) was allowed to vary. Which of the following relations regarding the initial rate of the reaction is correct?

(1) $R \propto \left(\frac{v}{V}\right)^m$ (2) $R \propto v^m$ (3) $R \propto v^{\frac{1}{m}}$ (4) $R \propto \left(\frac{v}{V}\right)^{\frac{1}{m}}$ (5) $R \propto V^m$

● For each of the questions **31** to **40**, one or more responses out of the four responses (*a*), (*b*), (*c*) and (*d*) given is/are correct. Select the correct response/responses. In accordance with the instructions given on your answer sheet, mark

(1) if only (*a*) and (*b*) are correct.
(2) if only (*b*) and (*c*) are correct.
(3) if only (*c*) and (*d*) are correct.
(4) if only (*d*) and (*a*) are correct.
(5) if **any other** number or combination of responses is correct.

Summary of above Instructions

(1)	(2)	(3)	(4)	(5)
Only (*a*) and (*b*) are correct	Only (*b*) and (*c*) are correct	Only (*c*) and (*d*) are correct	Only (*d*) and (*a*) are correct	**Any other** number or combination of responses is correct

31. Consider a titration between a weak acid (fixed volume) and a strong base. Which of the following is/are independent of the weak acid concentration?

(*a*) pH at the equivalence point.
(*b*) Volume of the strong base required to reach the end point.
(*c*) Dissociation constant of the weak acid.
(*d*) Value of $[H^+] \times [OH^-]$ of the solution in the titration flask.

32. Which of the following statements is/are **true** regarding the molecule given below?

$$\underset{\mathbf{a}}{CH_3}-\underset{\mathbf{b}}{C}\equiv\underset{\mathbf{c}}{C}-\underset{\mathbf{d}}{CHO}$$

(*a*) All four carbon atoms lie in the same plane.
(*b*) The angle between $C_{\mathbf{d}}-H$ and the $C_{\mathbf{d}}-C_{\mathbf{c}}$ bonds is approximately. 120°.
(*c*) Between $C_{\mathbf{b}}$ and $C_{\mathbf{c}}$, there are two σ-bonds and one π-bond.
(*d*) Between $C_{\mathbf{b}}$ and $C_{\mathbf{c}}$, there is one σ-bond and two π-bonds.

33. Which of the following statement/s is/are **true** with regard to the manufacture of Na_2CO_3?
(*a*) CO_2 is one of the raw materials used.
(*b*) The reaction between CO_2 and aqueous NaCl saturated with NH_3 is endothermic.
(*c*) The manufacturing process involves five stages.
(*d*) Most of the NH_3 used in the process can be recovered.

34. Temperature must be maintained at a constant value during the experimental determination of the order of an elementary reaction, because,
(*a*) the order of the reaction depends on temperature.
(*b*) the activation energy changes with temperature.
(*c*) the mechanism of the reaction changes with temperature.
(*d*) the rate constant changes with temperature.

35. Which of the following statement/s is/are **true** regarding ethene and ethyne?
(*a*) CaC_2 reacts with water to form ethyne.
(*b*) CaC_2 reacts with water to form ethene.
(*c*) Ethene reacts with ammoniacal $AgNO_3$ to give a precipitate.
(*d*) Ethyne reacts with ammoniacal Cu_2Cl_2 to give a precipitate.

36. Which of the following statement/s is/are **true** with regard to halogens?
(*a*) The boiling points of halogens increase down the group.
(*b*) Unlike other halogens, fluorine always has an oxidation state of (–1) except in F_2.
(*c*) All halogens are good reducing agents.
(*d*) Although fluorine is the most reactive of all the elements in the Periodic Table, it does not react with inert gases.

37. For the reaction $C(s)+CO_2(g) \rightleftharpoons 2\,CO(g)$ occurring in a closed rigid container, percentage yields of CO(g) at 700 °C and 800 °C are 60% and 80% respectively. Which of the following statement/s is/are **correct** regarding the above reaction?
(*a*) The reaction is endothermic.
(*b*) The reaction is exothermic.
(*c*) Reverse reaction is favoured by decreasing the temperature.
(*d*) Equilibrium can be shifted towards the reactants by removing C(s).

38. Cyclopropane ⟶ propene is an elementary reaction.
Which of the following statement/s is/are **correct** regarding the above reaction?
(*a*) Half life of the reaction depends on cyclopropane concentration.
(*b*) Rate of the reaction does not depend on propene concentration.
(*c*) The fraction of cyclopropane molecules having energy greater than the activation energy increases with increasing temperature.
(*d*) Reaction occurs *via* a bimolecular collision (molecularity = 2).

39. Which of the following statement/s is/are **true** regarding 3–hexene?
(*a*) Does not show geometric isomerism.
(*b*) Shows optical isomerism.
(*c*) The compound obtained when reacted with H_2/Pd does not show optical isomerism.
(*d*) The compound obtained when reacted with HBr shows optical isomerism.

40. Which of the following statements is/are **correct** with regard to the nitrogen cycle?

(*a*) N_2 in the atmosphere is fixed only by atmospheric and industrial fixation.
(*b*) N_2 is reduced during atmospheric fixation.
(*c*) N_2 is oxidized during industrial fixation.
(*d*) Nitrates and nitrites formed during atmospheric fixation are utilized by plants to make proteins when the rainfall deposit them on the ground.

- In question Nos. **41** to **50**, two statements are given in respect of each question. From the Table given below, select the response out of the responses (1), (2), (3), (4) and (5) that **best** fits the two statements and mark appropriately on your answer sheet.

Response	First Statement	Second Statement
(1)	True	True, and correctly explains the first statement
(2)	True	True, but does **not** explain the first statement correctly
(3)	True	False
(4)	False	True
(5)	False	False

	First Statement	Second statement
41.	$BaCO_3$ is more thermally stable than $MgCO_3$.	Polarizing power of group two cations decreases down the group.
42.	The lone pair of electrons on nitrogen in an amine has a lower tendency to form a bond with H^+, than the lone pair of electrons on oxygen in an alcohol.	Nitrogen is less electronegative than oxygen.
43.	A reaction at equilibrium can be driven forward (i.e. shift of equilibrium point to the right) by adding a catalyst.	The catalyst provides a pathway with a low activation energy only to the forward reaction.
44.	CO_3^{2-} and SO_3^{2-} ions have similar shapes.	Central atoms of both CO_3^{2-} and SO_3^{2-} have lone pairs of electrons.
45.	The boiling point of $CH_3CH_2CH_2OH$ is higher than the boiling points of CH_3CH_2CHO and CH_3COCH_3.	The carbon oxygen double bond is stronger than the carbon oxygen single bond.
46.	A reaction occurring spontaneously in an **isolated** system **always** has a negative Gibbs energy change.	A process in an **isolated** system cannot be changed from outside.
47.	Commonly used soap contain the sodium or potassium salts of fatty acids formed by the reaction of NaOH or KOH with oils and fats.	The reaction of an ester with aqueous NaOH or KOH gives the sodium or potassium salt of the carboxylic acid and the alcohol.
48.	C_6H_5Br does not react easily with NaOH to form C_6H_5OH.	The phenyl carbocation is very stable.
49.	When an aqueous solution of a weak acid is diluted, both the fraction of dissociated acid molecules and pH of the medium are increased.	Dissociation of weak acid molecules occur in such a way that the acid dissociation constant K_a remains constant.
50.	In the presence of sunlight CO_2 is fixed in green plants.	Increase of CO_2 level in the atmosphere cannot be controlled by green plants.

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ශ්‍රී ලංකා විභාග දෙපාර්තමේන්තුව
இலங்கைப் பரீட்சைத் திணைக்களம்

අ.පො.ස. (උ.පෙළ) විභාගය/ க.பொ.த. (உயர் தர)ப் பரீட்சை - 2018

විෂයය අංකය பாட இலக்கம்	02	විෂයය பாடம்	Chemistry

ලකුණු දීමේ පටිපාටිය/புள்ளி வழங்கும் திட்டம்

I පත්‍රය/பத்திரம் I

ප්‍රශ්න අංකය வினா இல.	පිළිතුරු අංකය விடை இல.	ප්‍රශ්න අංකය வினா இல.	පිළිතුරු අංකය விடை இல.	ප්‍රශ්න අංකය வினா இல.	පිළිතුරු අංකය விடை இல.	ප්‍රශ්න අංකය வினா இல.	පිළිතුරු අංකය விடை இல.	ප්‍රශ්න අංකය வினா இல.	පිළිතුරු අංකය விடை இல.
01.	04	11.	3	21.	2	31.	3	41.	1
02.	1 or 5 or both	12.	4	22.	4	32.	5	42.	4
03.	2	13.	1	23.	5	33.	3	43.	5
04.	5	14.	3	24.	4	34.	5	44.	5
05.	2	15.	3	25.	1	35.	4	45.	2
06.	1	16.	3	26.	3	36.	1 or 5 or both	46.	4
07.	4	17.	2	27.	1	37.	5	47.	1
08.	2	18.	4	28.	4	38.	2	48.	3
09.	5	19.	2	29.	3	39.	3	49.	1
10.	2	20.	2	30.	1	40.	5	50.	3

✪ විශේෂ උපදෙස්/ விசேட அறிவுறுத்தல் :

එක් පිළිතුරකට/ ஒரு சரியான விடைக்கு 01 ලකුණු බැගින්/புள்ளி வீதம்

මුළු ලකුණු/மொத்தப் புள்ளிகள் 1 × 50 = 50

PART A — STRUCTURED ESSAY

*Answer **all four** questions on this paper itself. (Each question carries **10** marks.)*

1. (*a*) State whether the following statements are **true** or **false**. (Reasons are **not** required.)

(i) The polarizability of halide ions increases with increasing size. ....True.....

(ii) The O—N—O bond angle of NO_2 is greater than that of NO_2^-. .....True.....

(iii) London dispersion forces among CCl_4 molecules are smaller than the London dispersion forces among SO_3 molecules. .....False.....

(iv) The shape of the HSO_4^- ion is trigonal bipyramidal. .....False.....

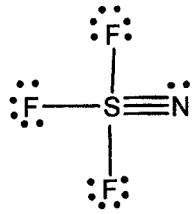
(v) All 3*d* atomic orbitals of an atom are represented by quantum numbers (n, l, m_l) 3, 2, 1. .....False.....

(vi) The addition of an electron to a gaseous phosphorus atom is an exothermic process whereas for a gaseous nitrogen atom it is endothermic. .....True.....

(✔ = True ✗ = False can be accepted.) **(04 marks x 6 = 24)**

1(a) = 24 marks

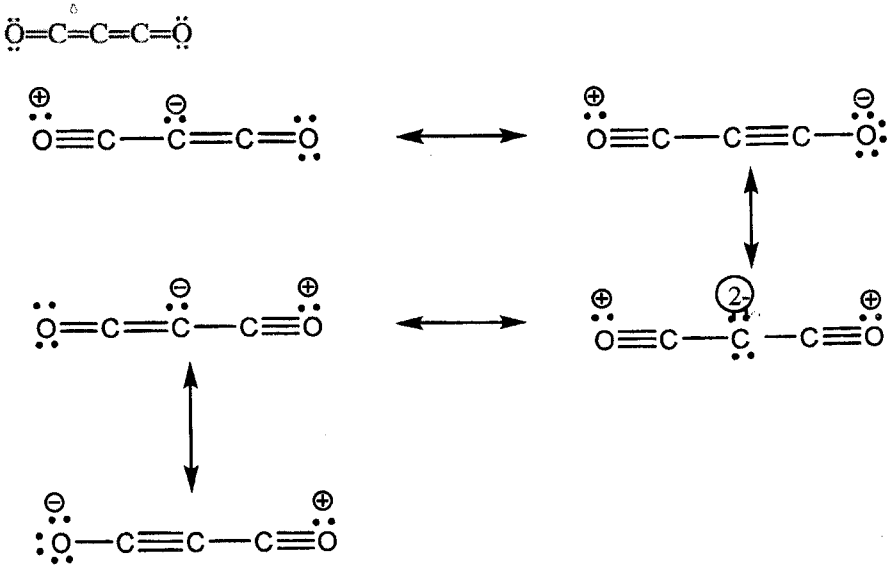
(*b*) (i) Draw the **most** acceptable Lewis structure for the molecule SF_3N.



(08)

(ii) The most stable Lewis structure for the molecule C_3O_2 (carbon suboxide) is shown below. Draw another **two** Lewis structures (resonance structures) for this molecule.

(**Note**: Marks will not be awarded for Lewis structures drawn with octet rule violated.)

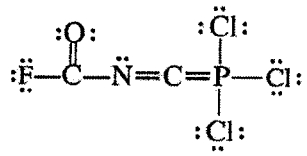


(any two) **(07 marks x 2 = 14)**

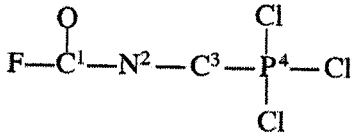
(resonance arrows are not required for award of marks)

(iii) Based on the Lewis structure given below, state the following regarding the C, N and P atoms given in the table below.

I. VSEPR pairs around the atom
II. electron pair geometry around the atom
III. shape around the atom
IV. hybridization of the atom



The atoms are numbered as follows:



		C^1	N^2	C^3	P^4
I	VSEPR pairs	3	3	2	4
II	Electron pair geometry	trigonal planar	trigonal planar	linear	tetrahedral
III	Shape	Trigonal planar	Angular / V	linear	tetrahedral
IV	Hybridization	sp^2	sp^2	sp	sp^3

(01 mark x 16 = 16)

(iv) Identify the atomic/hybrid orbitals involved in the formation of the following σ bonds in the Lewis structure given in part (iii) above. (Numbering of atoms is as in part (iii).)

I. $F—C^1$ F....$2p$ or sp^3...., C^1 ....sp^2....
II. $C^1—N^2$ C^1....sp^2...., N^2 ....sp^2....
III. $N^2—C^3$ N^2....sp^2...., C^3 ....sp....
IV. $C^3—P^4$ C^3....sp...., P^4 ....sp^3....
V. $P^4—Cl$ P^4....sp^3...., Cl ....$3p$ or sp^3....

(01 mark x 10 = 10)

(v) Identify the atomic orbitals involved in the formation of the following π bonds in the Lewis structure given in part (iii) above. (Numbering of atoms is as in part (iii).)

I. $N^2—C^3$ N^2....$2p$...., C^3 ....$2p$....
II. $C^3—P^4$ C^3....$2p$...., P^4 .....$3d$.... **(award 01 mark for any answer OR no answer)**

(01 mark x 4 = 04)

1(b) = 52 marks

(c) Arrange the following in the **increasing** order of the property indicated in parenthesis (Reasons are **not** required.)

(i) B, Na, P, Be, N (first ionization energy)

Na < B < Be < P < N **(08)**

(ii) NH_3, NOCl, NO_2Cl, NH_4^+, $F_3C—NC$ (electronegativity of nitrogen)

NH_3 < NH_4^+ < NOCl < NO_2Cl < CF_3NC **(08)**

Note: NH_3 < NOCl < NH_4^+ < NO_2Cl < CF_3NC (only for this year) **(08)**

(iii) Quantum numbers of electrons in an atom (n, l, m_l, m_s)

$\left(3, 1, 0, -\frac{1}{2}\right), \left(3, 0, 0, +\frac{1}{2}\right), \left(2, 0, 0, +\frac{1}{2}\right), \left(2, 1, +1, +\frac{1}{2}\right), \left(3, 2, -1, +\frac{1}{2}\right)$ (energy of electron)

(2,0,0,+1/2) < (2,1,1, +1/2) < (3,0,0, +1/2) < (3,1,0,-1/2) < (3,2,-1,+1/2) **(08)**

(08 marks x 3 = 24)

1(c) = 24 marks

2.(a) **X** is a *p*-block element in the Periodic Table. It exists as a diatomic gas. **X** exhibits a wide range of oxidation states. **Y** is the most common hydride of **X**. **Y** dissolves readily in water to give a basic solution. **Y** acts as an oxidizing agent, a reducing agent, an acid and a base. The diatomic gas of **X** is used in the manufacture of **Y**.

(i) Identify **X** and **Y**.

X – N or Nitrogen **(No marks for N_2)** **(05)**

Y – NH_3 or Ammonia **(05)**

(ii) The diatomic gas of **X** is generally considered as inert. Briefly explain.

N_2 contains a triple bond. **(03)**

Therefore, high bond dissociation energy **(03)**

(iii) Write the chemical formulae of **three** oxides of **X**, and indicate the oxidation state of **X** in each compound.

N_2O +1 NO +2 N_2O_3 +3

NO_2/ N_2O_4 +4 N_2O_5 +5 **(03 + 03 + 03)**

Note: Award mark for oxidation state only if formula is correct.

Mark distribution, formula (02), oxidation state (01)

(Any three can be accepted)

(iv) In each of the following instances, give a balanced chemical equation to indicate the action of **Y**.

I. **Y** as an oxidizing agent

$2Na(s) + 2NH_3(l) \longrightarrow 2NaNH_2(s) + H_2(g)$

$3Mg(s) + 2NH_3(l) \longrightarrow Mg_3N_2(s) + 3H_2(g)$

$6Li(s) + 2NH_3(l) \longrightarrow 2Li_3N(s) + 3H_2(g)$

(Any one) **(03)**

II. **Y** as a reducing agent

$3Cl_2(g) + 2NH_3(g) \longrightarrow N_2(g) + 6HCl(g)$

$3Cl_2(g) + 8NH_3(g) \longrightarrow N_2(g) + 6NH_4Cl(s)$

$3Cl_2(g) + NH_3(g) \longrightarrow NCl_3(l) + 3HCl(g)$

$3CuO(s) + 2NH_3(g) \longrightarrow N_2(g) + 3Cu(s) + 3H_2O(g)$

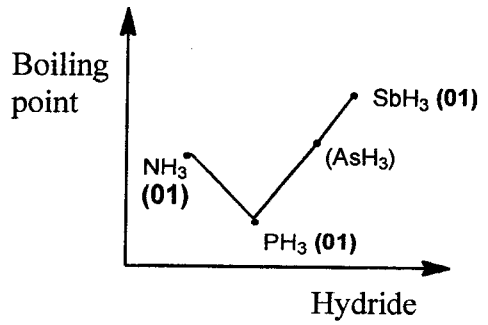
$3O_2(g) + 4NH_3(g) \longrightarrow 2N_2(g) + 6H_2O(l)$

$5O_2(g) + 4NH_3(g) \longrightarrow 4NO(g) + 6H_2O(l)$

(Any one) **(03)**

Note: Physical states are not required for award of marks.

(v) Consider the hydrides of the elements in the group to which **X** belongs, which are analogous to **Y**. Sketch the variation in boiling points of these hydrides (including **Y**) in the graph below. In your sketch indicate the hydrides using their chemical formulae. (**Note**: Values of boiling points are **not** required.)



(05)

Note: Shape (02). Shape needs to be correct for award of marks for labeling.
(i.e. Max. SbH_3; Min. PH_3; In between NH_3)

(vi) Give reasons for the variation in boiling points in part (v) above.

As molecular mass / size increases, boiling point increases. **(03)**

But with NH_3, boiling point is higher than expected because of H-bonding between NH_3 molecules. **(03)**

(vii) I. Write what you would observe when an excess of an aqueous solution of **Y** is added to a solution of $Al_2(SO_4)_3$.

white precipitate / white gelatinous precipitate ..... (03)

II. Write the chemical formula of the species that gives rise to your observation in part I above.

$Al(OH)_3$ ..... (03)

(viii) Give **one chemical** test to identify **Y**.

Test : Test with Nessler's reagent ..... (03)

Observation : Brown precipitate / Brown coloration ..... (03)

OR

Test with HCl vapour (03)

White fumes (03)

OR

Test with red litmus (03)

Red litmus turns blue (03)

OR

Add to a solution of Cu(II) ions (03)

Deep blue colour solution (03)

(ix) **Z** is an oxo-acid of **X** and a strong oxidizing agent.

I. Identify **Z**. HNO_3 **OR** Nitric acid ..... (03)

II. State the products obtained when hot concentrated **Z** reacts with sulphur.

$H_2SO_4(l)$, $NO_2(g)$, $H_2O(l)$ ..... (01+01+01)

Note: physical states are not required.

2(a) = 60 marks

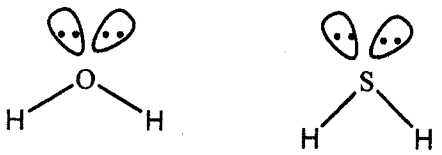
(*b*) **A** and **B** are compounds of two *p*-block elements that belong to the same group in the Periodic Table. **A** exists as a colourless, odourless liquid at room temperature and atmospheric pressure. It is also found in the gaseous and solid states. The solid state of **A** is less dense than its liquid state. Ionic and polar compounds are readily soluble in **A**.

B is a colourless gas at room temperature and atmospheric pressure. A filter paper moistened with lead acetate turns black on treatment with **B**.

(i) Identify **A** and **B**.

A - H_2O **B** - H_2S (04 + 04)

(ii) Sketch the shapes of **A** and **B** showing lone pairs of electrons where necessary.



A **B** **(03 + 03)**

(iii) Giving reasons, state whether **A** or **B** has the larger bond angle.

Oxygen is more electronegative than suphur. **(01)**
Therefore, bonding pairs of electrons are located closer to the oxygen atom in H_2O, than to the sulphur atom in H_2S. **(01)**
Therefore, repulsion of bonding electron pairs is greater in H_2O than in H_2S. **(01)**
Bond angle of **A**/H_2O is greater than bond angle of **B**/H_2S **(02)**

(iv) In each of the following instances, give a balanced chemical equation to indicate the action of **A**.

I. **A** as an acid: $H_2O(l) + NH_3(aq) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$ (**OR** $NH_4OH(aq)$) **(03)**
OR
$2Na(s) + 2H_2O(l) \longrightarrow 2NaOH(aq) + H_2(g)$
(**OR** any other metal that reacts with water liberating H_2)

(Note: $\longrightarrow$ accepted)

II. **A** as a base: $H_2O(l) + HCl(aq) \longrightarrow H_3O^+(aq) + Cl^-(aq)$ **OR** **(03)**
$H_2O(l) + CH_3COOH(aq) \longrightarrow H_3O^+(aq) + CH_3COO^-(aq)$

(v) Write the balanced chemical equation for the reaction of **B** with aqueous lead acetate.

$Pb(CH_3COO)_2(aq) + H_2S(g) \longrightarrow PbS(s) + 2CH_3COOH(aq)$ **(03)**
(**OR** $2CH_3COO^- + 2H^+$)

(vi) I. Write what you would observe when **A** and **B** are added **separately** to an acidified solution of $BiCl_3$.

with **A** (excess) - white precipitate / white solid/ white turbidity **(03)**

with **B** - black precipitate **(03)**

II. Write balanced chemical equations for your observations in part I above.

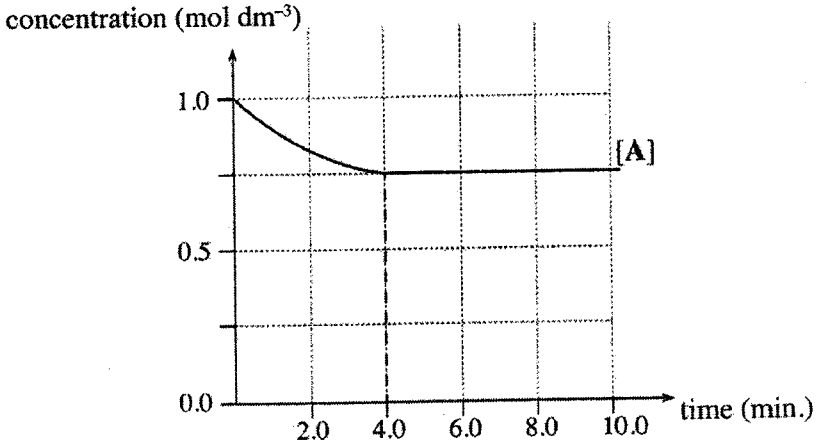
$BiCl_3(aq) + H_2O(l) \rightleftharpoons BiOCl(s) + 2HCl(aq)$ **(03)**
($\longrightarrow$ can be accepted.)

~~$2BiCl_3(aq) + 3H_2S(g) \longrightarrow Bi_2S_3(s) + 6HCl(aq)$~~ **(03)**

Note: Physical states are not required for parts (iv), (v), (vi)

2(b) = 40 marks

3. The reaction $A + B \rightleftharpoons 2C + D$ (elementary in both directions) was carried out at 25 °C. Initially, the reaction mixture was made by dissolving 0.10 mol of **A** and 0.10 mol of **B** in distilled water (total volume 100.00 cm^3). Variation in the concentration of **A** in this solution with time is shown in the graph.



(i) Calculate the amount of **A** (in moles) reacted during the first 4.0 minutes of the reaction.

Initial amount of A = 0.1 mol

Concentration of A after 4.0 min = 0.75 mol dm^{-3}

Amount of A reacted = (0.1 - 0.75)x100x10^{-3} mol **(04+01)**

= 0.025 mol. **(04+01)**

(ii) Would the rate of the forward reaction be less than the rate of the reverse reaction after 4.0 minutes? Explain your answer.

No **(05)**

Forward and reverse rates will be equal after 4.0 min.

OR The reaction has reached the equilibrium. **(05)**

(iii) Given that the rate constant of the forward reaction ($k_{forward}$) is 18.57 mol^{-1} dm^3 min^{-1}, calculate the initial rate of the forward reaction.

Rate of the forward reaction, $R_f = k[A][B]$ **(05)**

Initial rate of the forward reaction = 18.57 mol^{-1} dm^3 min^{-1} x 1.0 mol dm^{-3} x 1.0 mol dm^{-3} **(04+01)**

= 18.57 mol dm^{-3} min^{-1} **(04+01)**

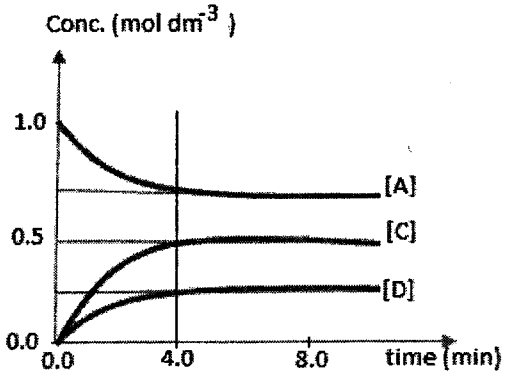
(iv) Calculate the concentrations of **C** and **D** at equilibrium.
Draw the relevant curves showing the variation of the concentrations of **C** and **D** with time in the above graph and **label** them.

Concentration of C at equilibrium = 2x 0.025 mol/(100.00 x 10^{-3} dm^3) **(02+01)**

= 0.50 mol dm^{-3} **(02+01)**

Concentration of D at equilibrium = 0.025 mol/(100.00x10^{-3} dm^3) **(02+01)**

= 0.25 mol dm^{-3} **(02+01)**



Curve C **(04)**

Curve D **(04)**

Note : Do not award marks if the curves do not become flat after 4.0 min, if the curves do not reach the respective concentrations at 4.0 min, if the curves for C and D are not labeled and if the curves do not start from zero.

(v) Write the expression for the equilibrium constant K_c of the above reaction and calculate its value.

(Equilibrium constant), $K_C = \frac{[C]^2\,[D]}{[A]\,[B]}$ **(05)**

$K_c = \frac{(0.5\ mol\ dm^{-3})^2(0.25\ mol\ dm^{-3})}{(0.75\ mol\ dm^{-3})(0.75\ mol\ dm^{-3})}$ **(04+01)**

$K_c = 1.11 \times 10^{-1}\ mol\ dm^{-3}$ **(04+01)**

(vi) Calculate the value of the rate constant ($k_{reverse}$) of the reverse reaction.

Using $K = \frac{k_f}{k_r}$, k_r can be calculated $k_r = \frac{18.57\ mol^{-1}dm^3min^{-1}}{1.11\times10^{-1}mol\ dm^{-3}}$ **(04+01)**

$k_r = 1.67 \times 10^2 mol^{-2}dm^6min^{-1}$ **(04+01)**

(vii) After reaching equilibrium, the volume of the solution was doubled by adding 100.00 cm^3 of distilled water. Predict the direction of the net reaction soon after doubling the volume of the solution, by means of a suitable calculation.

New concentrations,

[A] = 0.75/2 mol dm^{-3} , [B] = 0.75/2 mol dm^{-3}, [c] = 0.5/2 mol dm^{-3} , [D]= 0.25/2 mol dm^{-3}

Rate of forward reaction,

R_f = 18.57 mol^{-1} dm^3 min^{-1} (0.75/2 mol dm^{-3})^2 **(05+01)**

= 2.61 mol dm^{-3} min^{-1}

Rate of the reverse reaction,

R_r = 1.67x10^2 mol^{-2} dm^6 min^{-1} (0.5/2 mol dm^{-3})^2(0.25/2 mol dm^{-3}) **(05+01)**

= 1.30 mol dm^{-3} min^{-1}

$R_f > R_r$ Net reaction occurs in the forward direction. **(03)**

Alternate answer

$Q = \frac{(\frac{0.5}{2}\ mol\ dm^{-3})^2(\frac{0.25}{2}\ mol\ dm^{-3})}{(\frac{0.75}{2}\ mol\ dm^{-3})^2}$ **(05+01)**

$Q = 0.056\ mol\ dm^{-3}$ **(05+01)**

$Q < K$, therefore, the net reaction occurs in the forward direction. **(03)**

(viii) Consider that the above experiment was conducted at a temperature lower than 25 °C. How would this affect the rate of the reverse reaction? Explain your answer giving reasons.

Rate of the reverse reaction will decrease **(01)**

Because

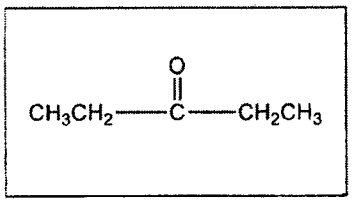
the fraction of molecules having sufficient energy to overcome the activation energy barrier decreases **(02)**

and

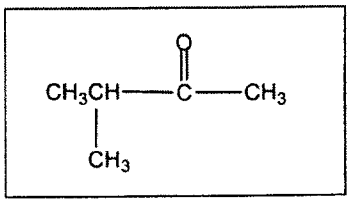
the collision rate decreases. **(02)**

Q3 = 100 marks

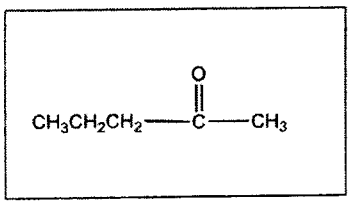
4. (*a*) (i) Compounds **A**, **B** and **C** are structural isomers of each other having the molecular formula $C_5H_{10}O$. All three compounds give yellow-orange precipitates with 2, 4–DNP. None of them give a silver mirror in the silver mirror test. When **A**, **B** and **C** were separately reacted with $NaBH_4$, compounds **D**, **E** and **F** respectively were obtained. Only **E** and **F** showed optical isomerism. When **B** and **C** were separately reacted with $CH_3CH_2CH_2MgBr$ followed by hydrolysis, compounds **G** and **H** respectively, were obtained. Only **G** showed optical isomerism. Draw the structures of **A**, **B**, **C**, **D**, **E**, **F**, **G** and **H** in the boxes given below. (It is **not** necessary to show stereoisomeric forms.)



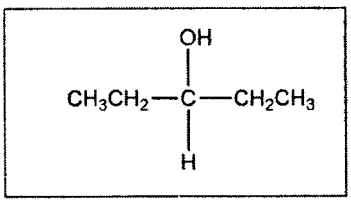
A



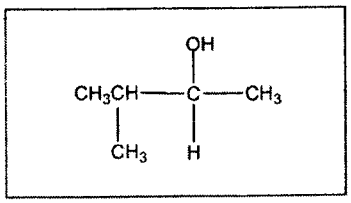
B



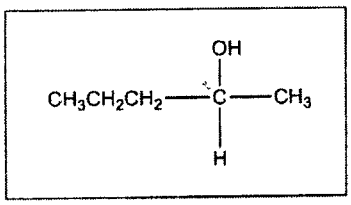
C



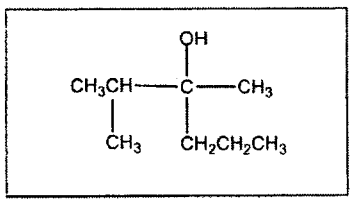
D



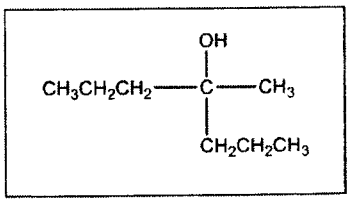
E



F



G



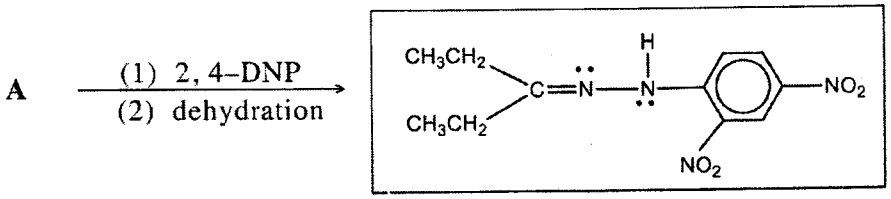
H

Note: C_3H_7 can be accepted in place of $CH_3CH_2CH_2$. C_2H_5 can be accepted in place of CH_3CH_2.

Note: A. B. C should be correct to award marks for D. E. F
B, C should be correct to award marks for G, H

(05 marks x 8 = 40 marks)

(ii) Draw the structure of the product of the following reaction.



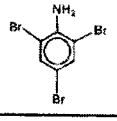
(05)

Note : Lone pairs are not necessary. Award mark if B or C is used instead of A, with the correct corresponding product.

4.(a): 45 marks

(*b*) Draw the structure of the **major** organic **product** in each of the following reactions.

(i) $C_6H_6 \xrightarrow[150\ °C]{H_2/\text{Raney Ni}}$  **(04)**

(ii) $C_6H_5-NH_2 \xrightarrow{Br_2 \text{ water}}$  **(04)**

(iii) $CH_3CHO \xrightarrow[(2)\ \text{dehydration}]{(1)\ \text{aqueous NaOH}}$ $CH_3CH{=}CH{-}C(=O){-}H$ **(04)**

(iv) $C_6H_5-\overset{\oplus}{N_2}\overset{\ominus}{Cl} \xrightarrow[\Delta]{H_3PO_2}$  **(04)**

(v) $C_2H_5CONH_2 \xrightarrow[\Delta]{\text{aqueous NaOH}}$ $C_2H_5{-}C(=O){-}O^-Na^+$ **(04)**

(vi) $CH_3CH{=}CH_2 \xrightarrow{\text{conc. } H_2SO_4}$ $CH_3{-}CH(OSO_3H){-}CH_3$ **(03)**

(vii) $CH_3COCl \xrightarrow{NH_3}$ $CH_3{-}C(=O){-}NH_2$ **(03)**

(viii) $C_2H_5CO_2H \xrightarrow{PCl_5}$ $C_2H_5{-}C(=O){-}Cl$ **(03)**

(ix) $C_2H_5OH \xrightarrow{H^+/KMnO_4}$ CH_3COOH **(03)**

(x) $C_2H_5COCH_3 \xrightarrow{HCN}$ $C_2H_5{-}C(OH)(CN){-}CH_3$ **(03)**

(i) strcture showing hydrogens on  can be accepted.

(iii) $CH_3CH{=}CHCHO$ can be accepted. No marks for $CH_3CH{=}CHCOH$

(iv)  can be acepted

(v) Charges on O and Na are not required for award of marks. No marks if given as O—Na

(vi) OSO_2OH can be accepted.

(vii) CH_3CONH_2 can be accepted.

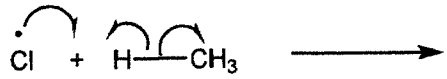
(viii) C_2H_5COCl can be accepted.

(ix) CH_3CO_2H can be accepted.

4(b): 35 marks



(04)

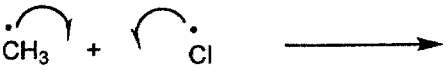


(08)



(08)

OR (for third step)



(02)

4.(c): 20 marks

Note: if no half arrow are drawn, deduct (01) mark once in each line.

Radical needs to be shown for award of marks.

Mark each step as an independent step.

PART B — ESSAY

Answer two questions only. (Each question carries **15** *marks.)*

5. (*a*) Consider the following reactions.

$$M(CO_3)_2 \cdot nH_2O(s) \rightarrow M(CO_3)_2(s) + nH_2O(g)$$

$$M(CO_3)_2(s) \rightleftharpoons MO_2(s) + 2CO_2(g)$$

A small amount (0.10 mol) of $M(CO_3)_2 \cdot nH_2O(s)$ is present in an evacuated rigid container of volume 0.08314 m^3. The temperature of the container was raised to 400 K. The metal carbonate, $M(CO_3)_2$ does not decompose at this temperature but the crystalline water evaporates completely. The pressure of the container was measured to be 1.60×10^4 Pa. Volume occupied by the solids is negligible.

Determine the value of 'n' in the formula $M(CO_3)_2 \cdot nH_2O(s)$.

$M(CO_3)_2.nH_2O$ (s) → $M(CO_3)_2$(s) + n H_2O(g)

$M(CO_3)_2$(s) ⇌ MO_2 (s) + $2CO_2$(g)

The amount of $M(CO_3)_2.nH_2O$ used = 0.10 mol

Water is completely evaporated.

Using PV=nRT, **(05)**

$$n_{H2O} = \frac{1.60\times10^4 Pa \times 0.08314\ m^3}{8.314\ J\ mol^{-1}K^{-1} \times 400K}$$ **(04+01)**

= 0.40 mol **(04+01)**

0.1 mol of $M(CO_3)_2.nH_2O$ (s) has generated 0.40 mol of H_2O. Therefore, n = 4 **(04+01)**

5 (a) = 20 marks

(*b*) The temperature of the above system was then increased to 800 K. It was observed that some amount of the solid metal carbonate is decomposed and is in equilibrium with the gas phase. The pressure of the container was measured to be 4.20×10^4 Pa.

(i) Calculate the partial pressure of water vapour in the container at 800 K.

Partial pressure of H_2O at 800 K,

$$P_{H2O} = \frac{n_{H2O}RT}{V}$$

$$= \frac{0.4\ mol \times 8.314\ J mol^{-1}K^{-1} \times 800\ K}{0.08314\ m^3}$$ **(04+01)**

= 3.20×10^4 Pa **(04+01)**

Alternate Answer 01

Total pressure at 800 K, $P_T = 4.20 \times 10^4$ Pa

Total number of moles $n_T = \frac{4.20 \times 10^4\ Pa \times 0.08314\ m^3}{8.314\ J\ mol^{-1}K^{-1} \times 800\ K}$ **(04+01)**

= 0.525 mol

Partial pressure of water = $P_{H2O} = P_T \times_{H2O}$

= 3.20×10^4 Pa **(04+01)**

Alternate Answer 02

Because V and n_{H2O} are constant, at 800 K,

partial pressure of water = $P_{H2O} = 2 \times 1.60 \times 10^4$ Pa **(04+01)**

= 3.20×10^4 Pa **(04+01)**

(ii) Calculate the partial pressure of CO_2 in the container at 800 K.

Partial pressure of CO_2 at 800K,

$P_{CO2} = P_{total} - P_{H2O}$

$= 4.2 \times 10^4$ Pa $- 3.2 \times 10^4$ Pa (04+01)

$= 1.00 \times 10^4$ Pa (04+01)

(iii) Write an expression for the pressure equilibrium constant, K_p for the decomposition of $M(CO_3)_2(s)$. Calculate K_p at 800 K.

$K_P = P^2_{CO2}$ (05)

$K_P = (1.0 \times 10^4 \text{ Pa})^2 = 1.00 \times 10^8 \text{ Pa}^2$ (04+01)

(iv) Calculate the molar percentage of the metal carbonate decomposed at 800 K.

Initial amount = 0.10 mol

Amount of CO_2 generated = n_{CO2}

$n_{CO2} = \frac{P_{CO2}V}{RT}$

$n_{CO2} = \frac{1.0 \times 10^4 Pa \times 0.08314\ m^3}{8.314\ J\ mol^{-1}K^{-1} \times 800K}$ **or** $\frac{3.2 \times 10^4 Pa}{1.0 \times 10^4\ Pa} = \frac{0.4}{n_{CO2}}$ (04+01)

$n_{CO2} = 0.125$ mol

Amount of $M(CO_3)_2$ decomposed = ½ amount of CO_2 generated.

mol % of $M(CO_3)_2$ decomposed $= \frac{0.0625\ mol}{0.10\ mol} \times 100$ (03)

$= 62.5$ % (02)

(v) Enthalpy change (ΔH) for the decomposition of the metal carbonate under the above conditions is 40.0 kJ mol^{-1}. Calculate the corresponding entropy change (ΔS).

System is at equilibrium, therefore $\Delta G = 0$. (05)

$\Delta S = \frac{\Delta H}{T}$

$\Delta S = \frac{40.0 \times 10^3\ J\ mol^{-1}}{800\ K}$ (04+01)

$\Delta S = 50.0$ J mol^{-1} K^{-1} **OR** 0.05 kJ mol^{-1} K^{-1} (04+01)

Note : ΔS^0 , ΔH^0 cannot be accepted.

(vi) Suggest **two** ways to drive the decomposition reaction of $M(CO_3)_2(s)$ in the forward direction.

Increase temperature (05)

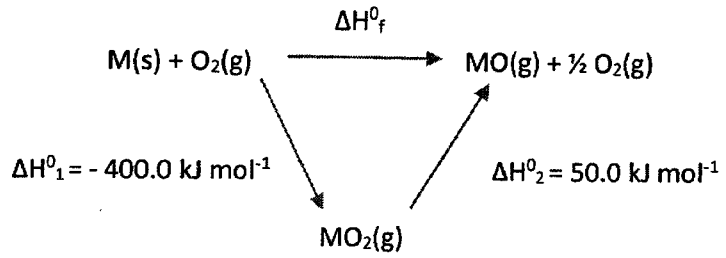
Remove CO_2 (05)

5 (b) = 65 marks

(c) With the aid of thermochemical cycles and the data given in the table, answer the following questions.

Species	Standard enthalpy of formation (ΔH_f°) (kJ mol^{-1})
M(s)	0.0
M(g)	800.0
O_2(g)	0.0
O(g)	249.2
MO_2(g)	−400.0

(i) Given that MO(g) + ½ O_2(g) → MO_2(g) ΔH° = −50.0 kJ mol^{-1}, calculate the standard enthalpy of formation of MO(g).



(02+02+02 =06)

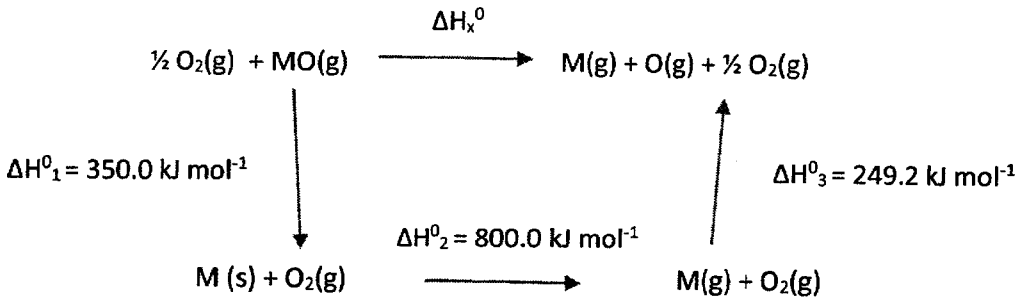
Note : To award marks for the cycle, reactions must be balanced and physical states must be given.

Standard formation enthalpy of MO(g), ΔH^0_f

ΔH^0_f = (-400.0 + 50.0) kJ mol^{-1} **(04+01)**

= -350.0 kJ mol^{-1} **(04+01)**

(ii) Calculate M–O bond dissociation enthalpy in MO(g).



(02+02+02 +02 = 08)

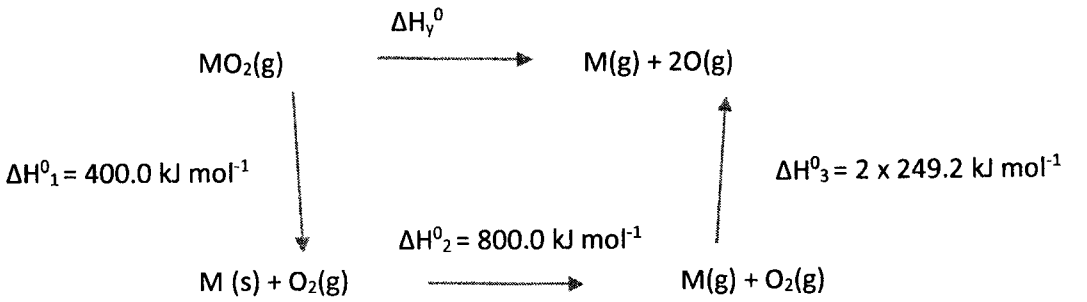
Note : To award marks for the cycle, reactions must be balanced and physical states must be given.

MO bond dissociation enthalpy change = ΔH_x^0

ΔH_x^0 = (350.0 + 800.0 + 249.2) kJ mol^{-1} **(04+01)**

= 1399.2 kJ mol^{-1} **(02+01)**

(iii) Calculate M–O bond dissociation enthalpy in $MO_2(g)$.



(02+02+02+02 =08)

Note : To award marks for the cycle, reactions must be balanced and physical states must be given.

$\Delta H_y^0 = (400.0 + 800.0 + 2 \times 249.2)$ kJ mol^{-1} **(04+01)**

$= 1698.4$ kJ mol^{-1}

MO bond dissociation energy in $MO_2 = \frac{1}{2} \Delta H_y^0 = 849.2$ kJ mol^{-1} **(04+01)**

(iv) By means of a suitable calculation, predict whether the reaction, $MO_2(g) \rightarrow MO(g) + \frac{1}{2} O_2(g)$ is spontaneous under standard conditions and 2000 K. Standard entropy change of this reaction is 30.0 J K^{-1} mol^{-1}.

$\Delta G^0 = \Delta H^0 - T\Delta S^0$ **(03)**

For the reaction, $MO_2(g) \longrightarrow MO(g) + \frac{1}{2} O_2(g)$ at 2000 K,

$\Delta G^0 = 50.0 \times 10^3$ J mol^{-1} – 2000 K x 30.0 J K^{-1} mol^{-1} **(04+01)**

$= -10000.0$ J mol^{-1} $= -10.0$ kJ mol^{-1} **(04+01)**

The given reaction is spontaneous at 2000K. **(02)**

Note : Standard states are required for award of marks.

5 (c) = 65 marks

6. (*a*) An experiment was carried out to determine the partition coefficient of iodine (I_2) between water (**A**) and an organic solvent (**B**) which form an immiscible liquid system.

20.00 cm^3 of **B** containing 'n' moles of I_2 is mixed with 20.00 cm^3 of **A** and allowed to reach equilibrium at room temperature.

The concentration of I_2 in phase **A** is determined by titrating a 5.00 cm^3 sample drawn from phase **A** with a 0.005 mol dm^{-3} solution of $Na_2S_2O_3$. The volume of $Na_2S_2O_3$ required to reach the end point was 22.00 cm^3. The concentration of I_2 in phase **B** was determined to be 0.040 mol dm^{-3}.

(i) Write the balanced chemical equation for the reaction between $Na_2S_2O_3$ and I_2.

$2Na_2S_2O_3 + I_2 \longrightarrow 2NaI + Na_2S_4O_6$

OR

$2 S_2O_3^{2-} + I_2 \longrightarrow 2 I^- + S_4O_6^{2-}$ **(10)**

(ii) Calculate the concentration of I_2 in phase **A**.

$$\text{Concentration of } I_2 \text{ in phase } A = \frac{22.00\ cm^3 \times 0.005\ mol\ dm^{-3}}{2 \times 5.0\ cm^3}$$ **(04+01)**

$= 0.011$ mol dm^{-3} **(04+01)**

(iii) Calculate the value of the partition coefficient, K_D where $K_D = \frac{[I_2]_B}{[I_2]_A}$.

Partition coefficient, $K_D = \frac{[I_2]_B}{[I_2]_A} = \frac{0.04\ mol\ dm^{-3}}{0.011\ mol\ dm^{-3}}$ **(04+01)**

(iv) Calculate the total number of moles of I_2 in the two phases **A** and **B**.

$K_D = 3.64$ **(04+01)**

Total number of moles of I_2

$n_{I_2} = 0.04\ mol\ dm^{-3} \times 20.0 \times 10^{-3} dm^3 + 0.011\ mol\ dm^{-3} \times 20.0 \times 10^{-3} dm^3$ **2 x (04+01)**

$= 1.02 \times 10^{-3}$ mol **(04+01)**

6 (a) = 45 marks

(*b*) The above experiment was repeated under the same conditions, that is, at the same temperature, using the same amount of I_2 and the same volumes, but with the addition of I^- ions to phase **A**. The system was then thoroughly shaken and allowed to reach equilibrium. The volume of 0.005 mol dm^{-3} $Na_2S_2O_3$ solution required to titrate the I_2 in a 5.00 cm^3 sample of phase **A** was 41.00 cm^3. The concentration of I_2 in phase **B** was then determined to be 0.030 mol dm^{-3}.

(i) Calculate the amount of I_2 (moles) expected in 5.00 cm^3 of phase **A**, based on the partition coefficient for the distribution of I_2 between the phases **A** and **B**.

Concentration of I_2 in phase A (when excess I^- is added)

$[I_2]_A = [I_2]_B/K_D$ **(05)**

$[I_2]_A = \frac{0.030\ mol\ dm^{-3}}{3.64}$ **(02+01)**

$= 8.242 \times 10^{-3}$ mol dm^{-3} **(01+01)**

The amount of I_2 in 5.00 cm^3 of phase A = n

n = 8.242×10^{-3} mol dm^{-3} x 5.00×10^{-3} dm^3 **(02+01)**

$= 4.121 \times 10^{-5}$ mol **(01+01)**

(ii) Calculate the amount (moles) of I_2 reacted with $Na_2S_2O_3$ in the above titration.

The amount of I_2 in 5.00 cm^3 of phase A, after the addition of iodide = n/

n/ = 0.005 mol dm^{-3} x 41.00 x 10^{-3} dm^3 x 0.5 **(04+01)**

$= 1.025 \times 10^{-4}$ mol (**or** 1.03×10^{-4} mol) **(04+01)**

(iii) Considering the different iodine species present in phase **A**, explain why the answers obtained in parts (*b*)(i) and (*b*)(ii) above are different.

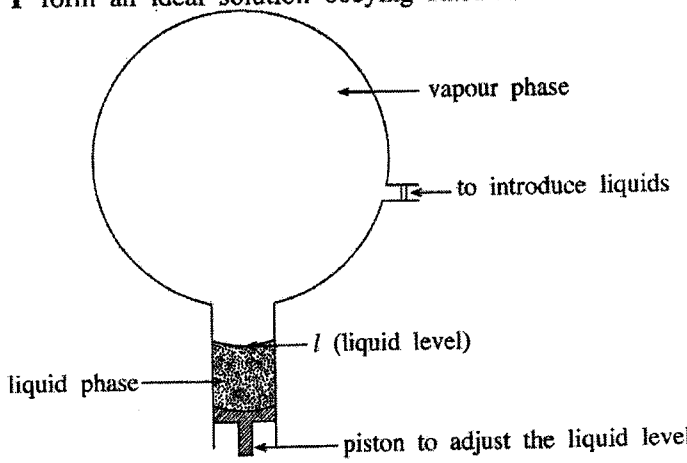
When I^- ions are added to the phase A, I_2 and I^- combine to form I_3^-. **(05)**

When phase A is titrated with $Na_2S_2O_3$, I_2 released from I_3^- is also reacted with $Na_2S_2O_3$.

Therefore, n/ > n. **(05)**

6 (b) = 35 marks

(c) Liquids **X** and **Y** form an ideal solution obeying Raoult's law.



Initially only liquid **X** was introduced in to an evacuated rigid container as shown in the figure.

Maintaining the liquid level at *l*, the system was allowed to reach equilibrium at 400 K. Th pressure of the container was measured to be 3.00×10^4 Pa. The volume of the vapour phas when the liquid level is at *l* was 4.157 dm^3. Then liquid **Y** was introduced in to the containe mixed with liquid **X** and the system was allowed to reach equilibrium at 400 K. The liqui level was maintained at *l*. The molar ratio of **X** : **Y** in the liquid phase was found to be 1 : 3 The pressure of the container was measured to be 5.00×10^4 Pa.

(i) What is the saturated vapour pressure of **X** at 400 K?

Saturated vapour pressure of X at 400K = 3.00×10^4 Pa. **(04+01)**

(ii) Calculate the mole fractions of **X** and **Y** in the liquid phase at equilibrium.

Mole fraction of X in the liquid phase = $\frac{1}{(1+3)}$ **(04+01)**

$= \frac{1}{4}$ or 0.25

Mole fraction of Y in the liquid phase = $\frac{3}{(1+3)}$ **(04+01)**

$= \frac{3}{4}$ *or* 0.75

(iii) Calculate the partial pressure of **X** at equilibrium after the addition of **Y**.

At equilibrium, $P_x = P_X^0 X_A$ **(05)**

$= 0.25 \times 3.0 \times 10^4$ Pa **(02+01)**

$= 7.5 \times 10^3$ Pa **(01+01)**

(iv) Calculate the partial pressure of **Y** at equilibrium.

$P_y = P_{total} - P_x$

$= 5.0 \times 10^4$ Pa $- 7.5 \times 10^3$ Pa **(02+01)**

$= 4.25 \times 10^4$ Pa **(01+01)**

(v) Calculate the saturated vapour pressure of **Y**.

Saturated vapour pressure of Y, $P_y^0 = \frac{P_y}{X_y}$

$$P_y^0 = \frac{4.25\times10^4 Pa}{0.75} \quad \textbf{(04+01)}$$

$$= 5.67 \times 10^4 Pa \quad \textbf{(04+01)}$$

(vi) Calculate the amounts (in moles) of **X** and **Y** in the vapour phase.

The amount of X in the gas phase, $n_x = P_xV/RT$

$$n_x = \frac{7.5\times10^3 Pa \times 4.157\times10^{-3} m^3}{8.314\ J\ mol^{-1}\ K^{-1} \times 400K} \quad \textbf{(04+01)}$$

$n_x = 9.38 \times 10^{-3}$ mol **(04+01)**

Similarly,

$$n_y = \frac{4.25\times10^4 Pa \times 4.157\times10^{-3} m^3}{8.314\ J\ mol^{-1}\ K^{-1} \times 400K} \quad \textbf{(04+01)}$$

$n_y = 5.31 \times 10^{-2}$ mol **(04+01)**

(vii) When a mixture of the liquids **X** and **Y** is subjected to fractional distillation, state which compound would distill out first from the fractional distillation column. Give reason/s for your answer.

Compound **Y** can be obtained first. **(05)**

Y is the more volatile compound or saturated vapour pressure of **Y** (P^0_y) is high. Therefore, its vapour comes out first from the fractional distillation column.

(05)

Note :To award marks for (vii) answers for P_x^0 and P_y^0 must have been calculated. Prediction must be according to the calculated P_x^0 and P_y^0 values.

6 (c) = 70 marks

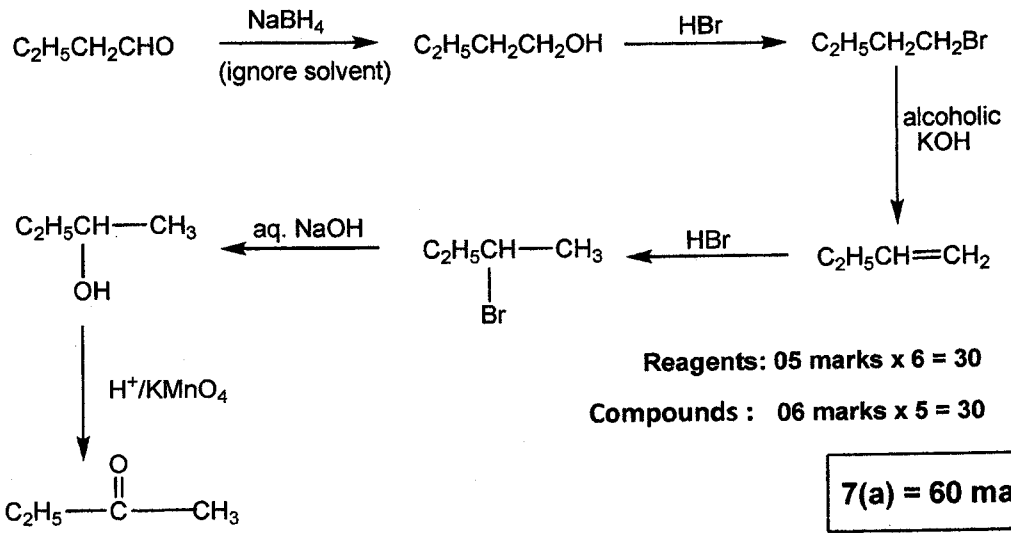
7. (*a*) Using only the chemicals given in the list, show how you would carry out the following conversion

$C_2H_5CH_2CHO \longrightarrow C_2H_5COCH_3$

List of chemicals
aqueous NaOH, HBr, alcoholic KOH, $NaBH_4$, $H^+/KMnO_4$

Your conversion should not exceed 7 steps.

7 (a)



Reagents: 05 marks x 6 = 30

Compounds : 06 marks x 5 = 30

7(a) = 60 marks

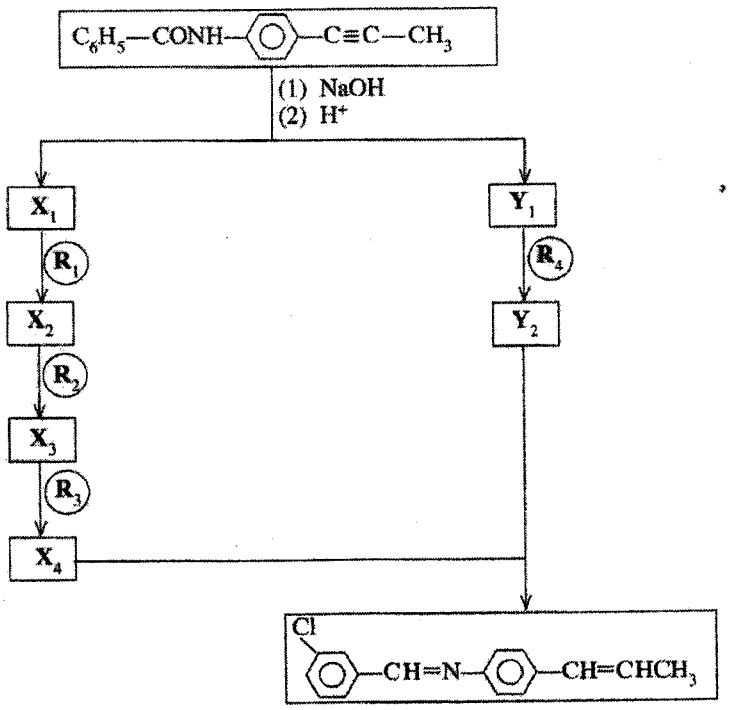
Note : Do not award 60 marks if there are more than seven steps. Do not award marks for $C_2H_5CH_2CHO$ and $C_2H_5COCH_3$.

Marking of partially correct answer

Mark from the beginning till an incorrect answer (reagent or product) is found. Mark from the end till an incorrect answer (reagent or product) is found. Add the marks. Do not award marks for any isolated correct steps in the middle.

To award marks for reagent, both reactant and product have to be correct.

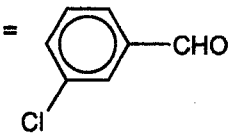
(*b*) Identify $\mathbf{R}_1$—$\mathbf{R}_4$ and $\mathbf{X}_1$—$\mathbf{X}_4$ and $\mathbf{Y}_1$, $\mathbf{Y}_2$ in order to complete the following reaction scheme

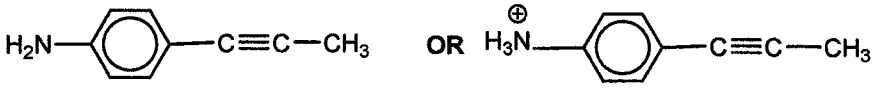


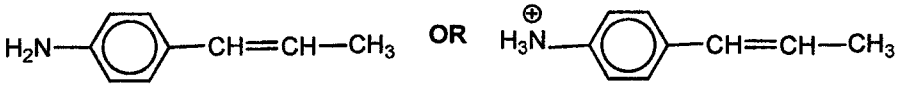
$X_1 =$ $C_6H_5CO_2H$ **(06)**

$X_2 =$ $C_6H_5CH_2OH$ **(06)**

$X_3 =$ C_6H_5CHO **(06)**

$X_4 =$  **(06)**

$Y_1 =$  **(06)**

$Y_2 =$  **(06)**

$R_1 =$ 1. $LiAlH_4$ 2. H_2O **OR** H_2O/H^+ **(06)**

$R_2 =$ PCC **(06)**

$R_3 =$ $FeCl_3/Cl_2$ **OR** Fe/Cl_2 **OR** Lewis acid/Cl_2 **(06)**

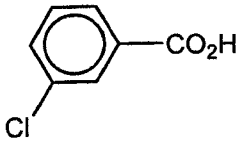
$R_4 =$ 1. NaOH 2. H_2/Pd/$BaSO_4$/quinoline **OR** **(06)**
1. NaOH 2. Lindlar/ H_2

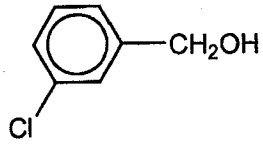
(Note : NaOH is not required for award of 06 marks) **(06 marks x 10 = 06)**

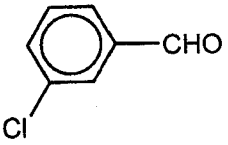
7(b) = 60 marks

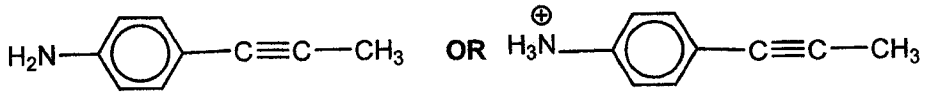
Alternative Pathway

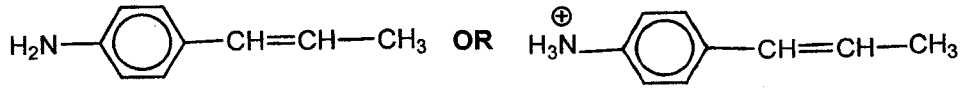
7 (b) X_1 = $C_6H_5CO_2H$ **(06)**

X_2 =  **(06)**

X_3 =  **(06)**

X_4 =  **(06)**

Y_1 =  **(06)**

Y_2 =  **(06)**

R_1 = $FeCl_3/Cl_2$ **OR** Fe/Cl_2 **OR** Lewis acid/Cl_2 **(06)**

R_2 = 1. $LiAlH_4$ 2. H_2O **OR** H_2O/H^+ **(06)**

R_3 = PCC **(06)**

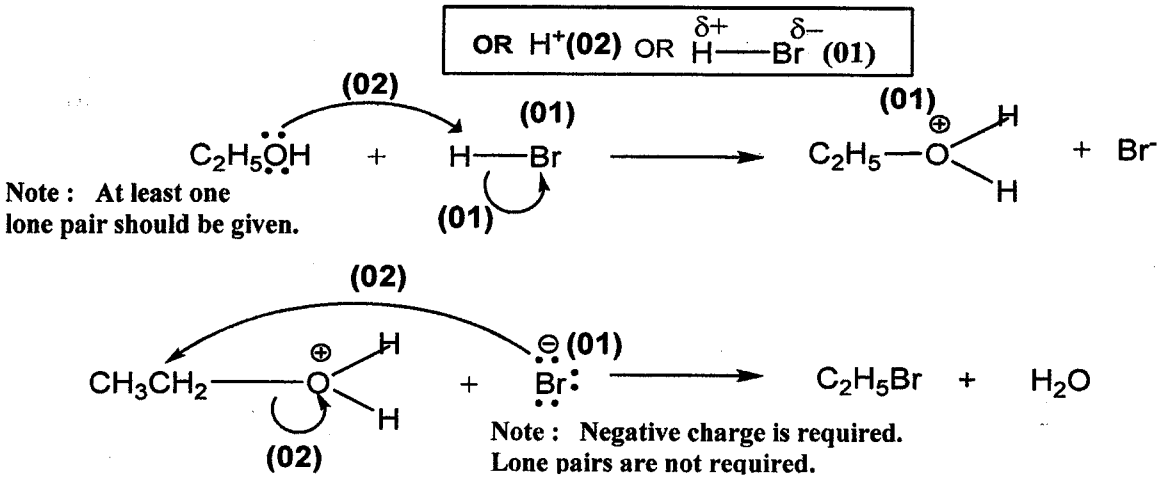
R_4 = 1. NaOH 2.H_2/Pd/$BaSO_4$/quinoline **OR** **(06)**
1. NaOH 2.Lindlar/H_2
(Note : NaOH is not required for award of 06 marks)

(06 marks x 10 = 60)

7(b) = 60 marks

(c) (i) Give the mechanism of the following reaction.

$$C_2H_5OH + HBr \longrightarrow C_2H_5Br + H_2O$$



(10 marks)

(ii) State whether the above reaction is a nucleophilic substitution reaction or an electrophilic substitution reaction. Identify the nucleophile or electrophile as appropriate.

(ii) Nucleophilic substitution, $Br^{\ominus}$

(02 + 02)

(iii) State giving reasons which of the two compounds, phenol (C_6H_5OH) or ethanol (C_2H_5OH) is more acidic.

$$C_2H_5OH + H_2O \rightleftharpoons C_2H_5O^{\ominus} + H_3O^{\oplus} \quad \textbf{(02)}$$

$$C_6H_5OH + H_2O \rightleftharpoons C_6H_5O^{\ominus} + H_3O^{\oplus} \quad \textbf{(02)}$$

Note: If H_2O is not included in the equations, only (01) per equation

- In the above equilibria the equilibrium point for phenol is more toward the right than ethanol. **(02)**
- This is because the stability of phenate ion relative to phenol is greater than the stability of the alkoxide relative to alcohol. **(02)**
- The phenate ion is more stable because its negative charge gets delocalized by resonance. **(02)**
- Resonance structures drawn **(02)**
- In alkoxide ion there is no such charge dispersion/ No resonance **(02)**
- Phenol is more acidic than ethanol. **(02)**

7(c) = 30 marks

PART C — ESSAY

Answer two questions only. (Each question carries **15** *marks.)*

8. (*a*) An aqueous solution **P** contains **two** cations and **two** anions. The following experiments were carried out to identify these cations and anions.

Cations

	Experiment	Observation
①	**P** was acidified with dilute HCl and H_2S was bubbled through the solution.	A clear solution was obtained.
②	The above solution was boiled till all the H_2S was removed. A few drops of conc. HNO_3 were added and the solution was heated further. The resulting solution was cooled and NH_4Cl/NH_4OH was added.	A brown precipitate (**Q**) was formed.
③	**Q** was removed by filtration and H_2S was bubbled through the filtrate.	A pale pink precipitate (**R**) was formed.
④	**R** was removed by filtration and the filtrate was boiled till all the H_2S was removed. $(NH_4)_2CO_3$ was added to the solution.	A clear solution was obtained.
⑤	Dilute NaOH was added to a **fresh portion** of **P**.	A dirty-green precipitate and a while precipitate were formed.

Experiments for precipitates **Q** and **R**:

	Experiment	Observation
⑥	**Q** was dissolved in dil. HNO_3 and a salicylic acid solution was added.	A light purple solution was obtained.
⑦	**R** was dissolved in dilute acid and dil. NaOH was added to the solution.	A white precipitate was formed. It turned brown on standing.

Anions

		Test	Observation
⑧	I.	$BaCl_2$ solution was added to **P**.	A white precipitate was formed.
	II.	The white precipitate was separated by filtration and dil. HCl was added to the precipitate.	The white precipitate was not dissolved.
⑨		Cl_2 water and chloroform were added to a portion of the filtrate from ⑧ II, and the mixture was throughly shaken.	Chloroform layer turned yellowish-brown.

(i) Identify the **two** cations and the **two** anions in solution **P**. (Reasons are **not** required.)

Cations: Fe^{2+} and Mn^{2+} **(10 + 10)**

Anions: SO_4^{2-} and Br^- **(08 + 07)**

Note: First correct anion (08), second anion (07)

(ii) Write the chemical formulae of the precipitates **Q** and **R**.

Q - $Fe(OH)_3$ **(10)**

R - MnS **(10)**

(iii) Give reasons for the following:

I. Removal of H_2S in experiment ② for cations.

- If H_2S is not removed MnS/FeS/ cations of group IV will also precipitate when NH_4OH/NH_4Cl solution is added. **(10)**

  OR
- H_2S can be oxidized to sulphur by conc. HNO_3. **(05)**
- A fine precipitate of sulphur would be formed in solution if H_2S is not removed. **(05)**

II. Heating with conc. HNO_3 in experiment ② for cations.

- K_{sp} of $Fe(OH)_2 > K_{sp}$ of $Fe(OH)_3$ **(05)**
- Therefore, Fe^{2+} needs to be converted to Fe^{3+} to be completely precipitated. **(05)**

  OR
- Conc. HNO_3 must be added to oxidize iron if present, to the ferric state. **(04)**
- If originally present, it would have been reduced by the H_2S to the ferrous ion. **(02)**
- Ferrous ion is not completely precipitated by NH_4OH/NH_4Cl solution.(will get a mixture of Fe^{2+} and Fe^{3+}) **(04)**

8(a): 75 marks

(*b*) The sample **X** contains lead, copper and an inert material. The following procedure was carried out to analyse lead and copper in **X**.

Procedure:

A mass of 0.285 g of **X** was dissolved in a slight excess of dil. HNO_3. A clear solution was obtained. A NaCl solution was added to the resulting clear solution. A white precipitate (**Y**) was formed. The precipitate was separated by filtration and the precipitate (**Y**) and filtrate (**Z**) were analysed separately.

Precipitate (Y)

The precipitate was dissolved in hot water. A solution of K_2CrO_4 was added in excess. A yellow precipitate was formed. The precipitate was separated by filtration and dissolved in dil. HNO_3. An orange coloured solution was obtained. Excess KI was added to this solution and the liberated I_2 was titrated with 0.100 mol dm^{-3} $Na_2S_2O_3$, with starch as the indicator. The volume of $Na_2S_2O_3$ required to reach the end point was 27.00 cm^3.

(Assume that the NO_3^- ions do **not** interfere with the titration.)

Filtrate (Z)

The filtrate was neutralized and excess KI was added to it. The liberated I_2 was titrated with 0.100 mol dm^{-3} $Na_2S_2O_3$, with starch as the indicator. The volume of $Na_2S_2O_3$ required to reach the end point was 15.00 cm^3.

(**Note**: Assume that the inert material was soluble in dil. HNO_3 and did **not** interfere with the experiment.)

(i) Calculate the mass percentages of lead and copper in **X**. Write balanced chemical equations where relevant.

Determination of Cu

$2Cu^{2+} + 4I^- \rightarrow I_2 + 2CuI$ ----------(1) **(05)**

OR

$2Cu^{2+} + 2I^- \rightarrow 2Cu^+ + I_2$ ----------(1) **(05)**

$I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$ ----------(2) **(05)**

From (1) and (2) $Cu^{2+} \equiv S_2O_3^{2-}$ **OR** Identification of correct stoichiometry **(02)**

Moles of $S_2O_3^{2-}$ $= \frac{0.10}{1000} \times 15.0$ **(03)**

Therefore, moles of Cu^{2+} $= \frac{0.10}{1000} \times 15.0$ **(03)**

Mass of Cu $= \frac{0.10}{1000} \times 15.0 \times 63.5$ **(03)**

$= 0.095$ g **(03)**

Therefore, %Cu $= \frac{0.095}{0.285} \times 100$ **(03)**

$=$ **33.4%** **(03)**

(30 marks)

Determination of Pb

$Cr_2O_7^{2-} + 6I^- + 14H^+ \rightarrow 2Cr^{3+} + 3I_2 + 7H_2O$ ----------(3) **(07)**

$I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$ ----------(4)

From (3) + 3x (4) $Cr_2O_7^{2-} \equiv 6S_2O_3^{2-}$ **OR** Identification of correct stoichiometry **(03)**

Moles of $S_2O_3^{2-}$ $= \frac{0.10}{1000} \times 27.0$ **(03)**

Moles of $Cr_2O_7^{2-}$ $= \frac{1}{6} \times \frac{0.10}{1000} \times 27.0$ **(03)**

$2CrO_4^{2-} + 2H^+ \rightarrow Cr_2O_7^{2-} + H_2O$ **(03)**

Therefore, moles of Cr $= 2 \times \frac{1}{6} \times \frac{0.10}{1000} \times 27.0$ **(03)**

Yellow precipitate is $PbCrO_4$ **(03)**

Therefore, moles of Pb $= 2 \times \frac{1}{6} \times \frac{0.10}{1000} \times 27.0$ **(03)**

Therefore, mass of Pb $= 2 \times \frac{1}{6} \times \frac{0.10}{1000} \times 27.0 \times 207$ **(03)**

$= 0.186$ g **(03)**

Therefore, %Pb $= \frac{0.186}{0.285} \times 100$ **(03)**

= 65.3% **(03)**

(40 marks)

Alternate method

Determination of Pb

$2CrO_4^{2-} + 6I^- + 16H^+ \rightarrow 2Cr^{3+} + 3I_2 + 4H_2O$ ----------(3)

$I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$ ----------(4)

OR

$CrO_4^{2-} + 8H^+ + 3e \rightarrow Cr^{3+} + 4H_2O$

$2I^- \rightarrow I_2 + 2e$

From equations $CrO_4^{2-} \equiv 3S_2O_3^{2-}$ **OR** Identification of correct stoichiometry **(03)**

Moles of $S_2O_3^{2-}$ $= \frac{0.10}{1000} \times 27.0$ **(03)**

Moles of I_2 $= \frac{1}{2} \times \frac{0.10}{1000} \times 27.0$ **(03)**

Moles of Cr^{3+} $= \frac{2}{3} \times \frac{1}{2} \times \frac{0.10}{1000} \times 27.0$ **(03)**

$= 9 \times 10^{-4}$

Therefore, moles of $PbCrO_4$ $= \frac{2}{3} \times \frac{1}{2} \times \frac{0.10}{1000} \times 27.0 = 9 \times 10^{-4}$ **(03)**

Therefore, moles of Pb $= \frac{2}{3} \times \frac{1}{2} \times \frac{0.10}{1000} \times 27.0 = 9 \times 10^{-4}$ **(03)**

Therefore, mass of Pb $= 9 \times 10^{-4} \times 207$ g **(03)**

$= 0.186$ g **(03)**

Therefore, %Pb $= \frac{0.186}{0.285} \times 100$ **(03)**

= 65.3% **(03)**

(30 marks)

(ii) What is the colour change at the end point in the titration carried out in the analysis of precipitate **Y**?
(Cu = 63.5, Pb = 207)

Blue → Green **(05)**

8(b): 75 marks

9. (*a*) The following questions are based on the environment and related issues.

(i) Identify **three** greenhouse gases that contribute to global warming. State **two** consequences of global warming.

Greenhouse gases that contribute to global warming.

CO_2, NO_x, N_2O, O_3, CFC, methane, volatile hydrocarbons **(03 + 03 + 03)**

Consequences:

- Melting of polar ice caps
- Change in weather patterns
- Drying up of freshwater reservoirs
- Sinking of low lying countries due to thermal expansion of sea water/ sea level rise
- desertification
- loss of soil moisture
- changes in biodiversity
- decrease in dissolved oxygen content
- increase in populations of certain insects

(Any two) **(03 + 03)**

(ii) Global environmental issues caused by coal power plants are well known. Identify **one** such issue that contributes **significantly** to change in certain water quality parameters in rivers and lakes.

Acid rain **(03)**

(iii) Name the chemical species responsible for the environmental issue identified in (ii) above and state **three** water quality parameters that are likely to be affected by this issue.

SO_2/ SO_3 / H_2SO_3 / H_2SO_4 **(03)**

Water parameters affected:

- pH (decreases) / acidity (increases)
- Salinity (increases)
- Concentration of heavy metals (increases)
- Hardness (increases)
- Conductivity (increases)

(Any three) **(03 + 03 + 03)**

(iv) Identify **two** environmental issues that change (increase or decrease) the ozone level in the atmosphere and explain briefly how these changes take place with the aid of balanced chemical equations.

photochemical smog (Ozone **increases)** **(03)**

How

Vehicle emissions contain NO_x **(03)**

$NO_2(g) \xrightarrow{h\nu} NO(g) + O(g)$ **(03)**

$M + O(g) + O_2(g) \longrightarrow O_3(g) + \overset{*}{M}$ **(03)**

(**M** = third body)

ozone layer depletion (Ozone **decreases)** **(03)**

How

Ozone destroyed by free radicals $(\dot{X})$ (e.g. $\dot{H}$, $\dot{N}O$, $\dot{O}H$, $\dot{C}l$) **(03)**
which act as a catalyst.

$O_3(g) + \dot{X}(g) \longrightarrow \dot{O}X(g) + O_2(g)$ ---------(1) **(03)**

$O_2(g) \longrightarrow 2O(g)$ ---------(2) **(03)**

$\dot{X}O(g) + O(g) \longrightarrow \dot{X}(g) + O_2(g)$ ---------(3) **(03)**

(1)x2 + (2) + (3)x2

$2O_3(g) \longrightarrow 3O_2(g)$ **(03)**

(v) I. "Most of the harmful gases in vehicle exhausts are converted to relatively harmless gases by catalytic converters." Briefly explain this statement.

Catalytic converters convert

- $NO(g)$ formed to $N_2(g)$ **(03)**
- $CO(g)$ formed to $CO_2(g)$ **(03)**
- Unburnt or partially burnt hydrocarbons to $CO_2(g) + H_2O(g)$ **(03)**

II. Name the harmful gas (except CO_2) that is not converted to a less harmful gas by the catalytic converter. State briefly how this harmful gas is formed in the vehicle engine.

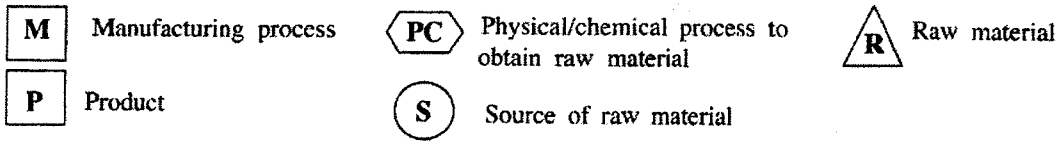
SO_2 **(03)**

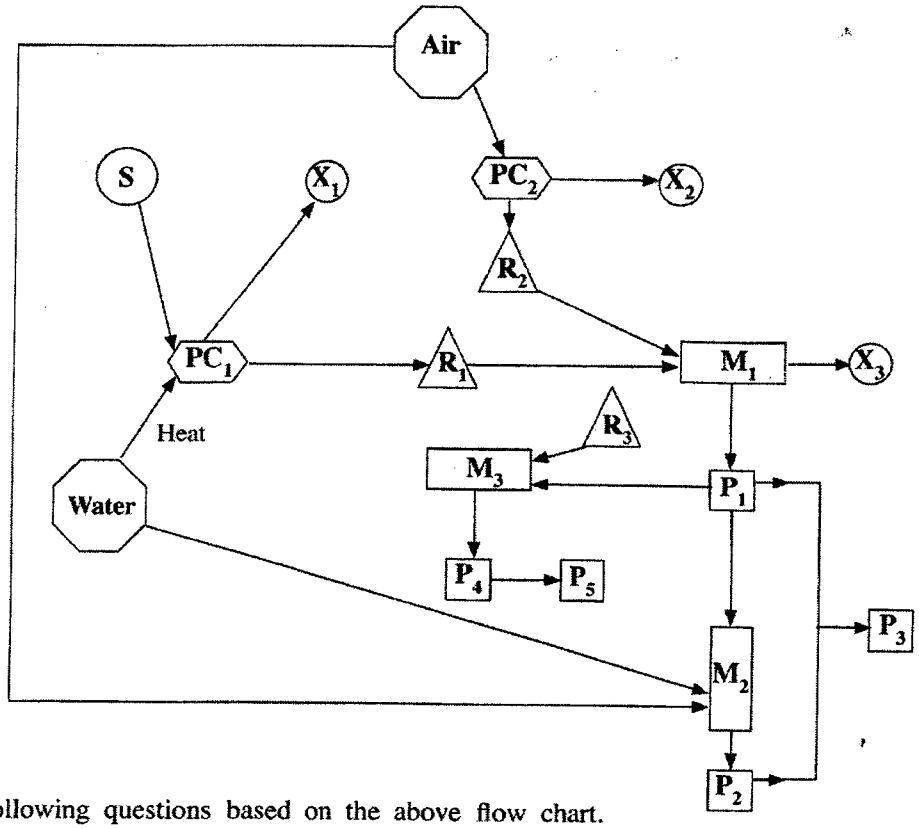
Certain fossil fuels contain sulphur. **(02)**

Burning of sulphur produces SO_2. **(01)**

9(a): 75 marks

(*b*) The flow chart given below shows the production of two important compounds P_1 and P_2 and three other important compounds P_3, P_4 and P_5 derived from them. P_1 is used as a raw material in the manufacture of Na_2CO_3. P_3 can be manufactured by the reaction between P_1 and P_2. P_3 is used as a fertilizer and as an explosive. P_1 is also used in the manufacture of P_4 which is a widely used fertilizer. P_4 is used to synthesize an important thermosetting polymer P_5.





Answer the following questions based on the above flow chart.

(i) Identify P_1, P_2, P_3, P_4 and P_5

$P_1 = NH_3$ **(03)**

$P_2 = HNO_3$ **(03)**

$P_3 = NH_4NO_3$ **(03)**

$P_4 = urea/ CO(NH_2)_2$ **(03)**

$P_5 = urea\text{-}formadehyde$ **(03)**

(ii) Identify $\mathbf{R_1}$, $\mathbf{R_2}$ and $\mathbf{R_3}$

$R_1 = H_2$ **(03)**

$R_2 = N_2$ **(03)**

$R_3 = CO_2$ **(03)**

(iii) Identify $\mathbf{X_1}$, $\mathbf{X_2}$ and $\mathbf{X_3}$

$X_1 = CO / CO_2$ **(03)**

$X_2 = O_2$ **(03)**

$X_3 = N_2 + H_2$ **(Both N_2 and H_2 must be present for the award of marks)** **(03)**

(iv) Identify S.

S = natural gas /CH_4 **OR** naptha/ (C_6H_{14}) **OR** coal / (carbon) **(02)**

(v) Briefly state the processes taking place in $\mathbf{PC_1}$ and $\mathbf{PC_2}$ giving balanced chemical equations where applicable.

PC_1: $C_6H_{14}(s) + 6H_2O(g) \longrightarrow 6CO(g) + 13H_2(g)$ **(02)**

OR

$CH_4(g) + H_2O(g) \longrightarrow CO(g) + 3H_2(g)$ **(02)**

PC_2: fractional distillation of liquid air **(02)**

(vi) Identify manufacturing processes $\mathbf{M_1}$, $\mathbf{M_2}$ and $\mathbf{M_3}$. (e.g. contact process or manufacture of H_2SO_4.)

M_1 = Haber process **OR** manufacture of NH_3 **(02)**

M_2 = Ostwald process **OR** manufacture of HNO_3 **(02)**

M_3 = manufacture of urea **(02)**

(vii) Give balanced chemical equations with appropriate conditions, for reactions taking place in $\mathbf{M_1}$, $\mathbf{M_2}$ and $\mathbf{M_3}$.

$\mathbf{M_1}$: $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ **(01)**

(200- 250) atm

(400 - 450) °C **(01)**

Fe (catalyst) **(01)**

K_2O and Al_2O_3 (promoters) **(01)**

(02)

$\mathbf{M_2}$: $4NH_3(g) + 5O_2(g) \xrightleftharpoons[\text{Pt or Pt-Rh(catalyst) } \textbf{(01)}]{(800 - 900)\ ^\circ C\ \textbf{(01)}} 4NO(g) + 6H_2O(g)$ **(02)**

$2NO(g) + O_2(g) \xrightarrow{150\ ^\circ C\ \textbf{(01)}} 2NO_2$ **(02)**

$4NO_2(g) + 2H_2O(l) + O_2(g) \longrightarrow 4HNO_3(aq)$ **(02)**

$\mathbf{M_3}$: $2NH_3(l) + CO_2(l) \rightleftharpoons NH_2COONH_4(s)$ **(02)**

(180-210) °C **(01)**

(30-150) atm **(01)**

$NH_2COONH_4(s) \longrightarrow (NH_2)_2CO(aq) + H_2O(l)$ **(02)**

↓ Concentrate by evaporation **(01)**

$(NH_2)_2CO(s)$ **(02)**

Note: Physical states are not required.

(viii) I. Give one use of each compound $\mathbf{P_1}$ and $\mathbf{P_2}$ other than those mentioned above.

P_1:

- Neutralizing acidic constituents in industry / emissions/ effluents / water treatment plants
- In stack emission control systems to neutralize sulphur oxides from combustion of sulphur-containing fuels
- As a refrigerant
- In the rubber industry / for the stabilization of natural and/or synthetic latex / to prevent premature coagulation
- In the paint industry

(Any one) **(02)**

P_2:

- To manufacture nitrates **OR**
  $NaNO_3$ - meat preservative **OR**
  $AgNO_3$ - prepare photographic films and paper
- For the preparation of aqua regia
- Used to clean soldering surfaces

(Any one) **(02)**

II. Give one use of R_1 in the manufacturing process P_1 other than being used as a raw material.

As a fuel **OR** to heat the system (to 450 °C) **(02)**

9(b): 75 marks

10. *(a)* **A** and **B** are **complex ions**, (i.e. metal ion and ligands coordinated to it) with an octahedral geometry. They have the same atomic composition of $MnC_5H_3N_6$. In each complex ion, **two** types of ligands are coordinated to the metal ion. When an aqueous solution containing **A** is treated with a potassium salt, the **coordination compound C** is formed. **C** gives four ions in aqueous solution. When an aqueous solution containing **B** is treated with a potassium salt the **coordination compound D** is formed. **D** gives three ions in aqueous solution. Both **C** and **D** have an octahedral geometry.

(**Note**: The oxidation states of manganese in **A** and **B** do not change on treatment with the potassium salt).

(i) Identify the ligands coordinated to manganese in **A** and **B**.

CN^- and NH_3 **(05 + 05)**

(ii) Give the structures of **A**, **B**, **C** and **D**.

A: $[Mn(CN)_5(NH_3)]^{3-}$ **OR** $[Mn(NH_3)(CN)_5]^{3-}$ **(10)**

B: $[Mn(CN)_5(NH_3)]^{2-}$ **OR** $[Mn(NH_3)(CN)_5]^{2-}$ **(10)**

C: $K_3[Mn(CN)_5(NH_3)]$ **OR** $K_3[Mn(NH_3)(CN)_5]$ **(15)**

D: $K_2[Mn(CN)_5(NH_3)]$ **OR** $K_2[Mn(NH_3)(CN)_5]$ **(15)**

(iii) Write the electronic configurations of the manganese ions in **A** and **B**.

A, oxidation state of Mn = +2
Therefore, electronic configuration of Mn in **A** is, $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$ **(03)**

B, oxidation state of Mn = +3
Therefore, electronic configuration of Mn in **B** is, $1s^2 2s^2 2p^6 3s^2 3p^6 3d^4$ **(02)**

(iv) Write the IUPAC names of **C** and **D**.

C potassium amminepentacyanidomanganate(II) **(05)**

D potassium amminepentacyanidomanganate(III) **(05)**

Note : If spelling is incorrect do not award marks.

10(a): 75 marks

(*b*) (i) I. Write the reduction half reaction corresponding to the electrode, $Ag(s)\,|\,AgCl(s)\,|\,Cl^-(aq)$.

$AgCl(s) + e \longrightarrow Ag(s) + Cl^-(aq)$ **(05)**

($\rightleftharpoons$ accepted). Physical states are required.

II. State whether the electrode potential of $Ag(s)\,|\,AgCl(s)\,|\,Cl^-(aq)$ depends on the Ag^+ concentration in the solution. Explain your answer.

No. **(05)**

$Ag^+(aq)$ does not appear in the electrode reaction (half reaction). **(05)**

(ii) Consider the following reaction.

$$Fe(s) + 2H^+(aq) + \frac{1}{2}O_2(g) \longrightarrow Fe^{2+}(aq) + H_2O(l)$$

I. Write the oxidation and reduction half reactions relevant to the above reaction.

$Fe(s) \longrightarrow Fe^{2+}(aq) + 2e$ (oxidation half reaction) **(08)**

$\frac{1}{2}O_2(g) + 2H^+(aq) + 2e \longrightarrow H_2O(l)$ (reduction half reaction) **(08)**

($\rightleftharpoons$ accepted) Physical states are required.

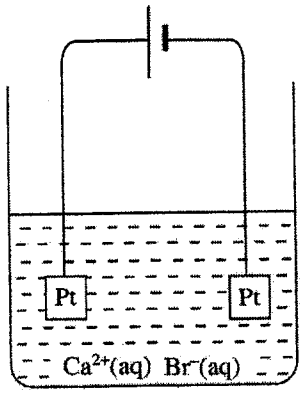
II. Given that the above reaction is the cell reaction of an electrochemical cell, determine the standard electromotive force of the cell.

$E^\circ_{Fe^{2+}(aq)/Fe(s)} = -0.44\ V$ $\qquad$ $E^\circ_{H^+(aq)/O_2(g)/H_2O(l)} = 1.23\ V$

Standard cell potential = 1.23V – (-0.44V) **OR** (1.23 – (-0.44))V **(01+01) + (01+01)**

= 1.67 V **(04+01)**

(iii) A constant current of 100 mA was passed through 100.00 cm^3 of a 0.10 mol dm^{-3} aqueous $CaBr_2$ solution as shown in the diagram. The temperature of the system was maintained at 25 °C.



I. Write the oxidation and reduction reactions that take place at the electrodes.

Oxidation half reaction,

$2\ Br^-(aq) \longrightarrow Br_2(g) + 2e$ **OR** $2Br^-(aq) \longrightarrow Br_2(l) + 2e$ **(05)**

Reduction half reaction,

$2H_2O(l) + 2e \longrightarrow H_2(g) + 2OH^-(aq)$ **(05)**

($\rightleftharpoons$ accepted). Physical states are required.

II. Calculate the time taken for the commencement of precipitation of $Ca(OH)_2(s)$. Solubility product of $Ca(OH)_2$ at 25 °C is 1.0×10^{-5} $mol^3\ dm^{-9}$. Neglect the ionization of water. Assume that the volume of the aqueous phase remains constant.

$K_{sp} = [Ca^{2+}(aq)][OH^-(aq)]^2$ **(05)**

Required concentration of OH^- to start precipitation of $Ca(OH)_2 = [OH^-]$

$$[OH^-] = \sqrt{\frac{1.0\times10^{-5} mol^{\,3} dm^{-9}}{0.1\ mol\ dm^{-3}}}$$ **OR** 1.0×10^{-2} mol dm^{-3} **(04+01)**

The amount of OH^- required to provide the above concentration = n_{OH^-}

$n_{OH^-} = 1.0 \times 10^{-2}$ mol $dm^{-3} \times 100 \times 10^{-3}$ dm^3 **OR** 1.0×10^{-3} mol **(04+01)**

Amount of charge, that must be passed through the solution, Q,

$Q = 1.0 \times 10^{-3}$ mol $\times$ 96500 C mol^{-1} **OR** 96.5 C **(04+01)**

Time required to pass the charge Q, when the current flow is 100 mA, = t

$$t = \frac{96.5\ C}{100 \times 10^{-3} Cs^{-1}}$$ **OR** $965\ s$ **OR** $16.08\ min$ **(04+01)**

(For the Faraday constant, a value between 96500 ± 100 C mol^{-1} is accepted, If the symbol F is used for the Faraday constant, and t is calculated using F, full marks can be awarded. t = 16.08 min **OR** t = 16 min accepted.)

10 (b) = 75 marks

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